

POWER BI



DATA
CIVILISATION

DATA VISUALISATION

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1. WHAT IS DATA VISUALISATION?



WHY IS DATA VISUALISATION IMPORTANT?

- ADDS A **NEW DYNAMIC** TO YOUR RESULTS AND FINDINGS
- TARGETS A DIFFERENT AUDIENCE AND IS **VERY POWERFUL** IN SHOWING YOUR FINDINGS
- A LOT **MORE EXCITING** TO SEE THAN REPORTS GENERATED FROM SQL

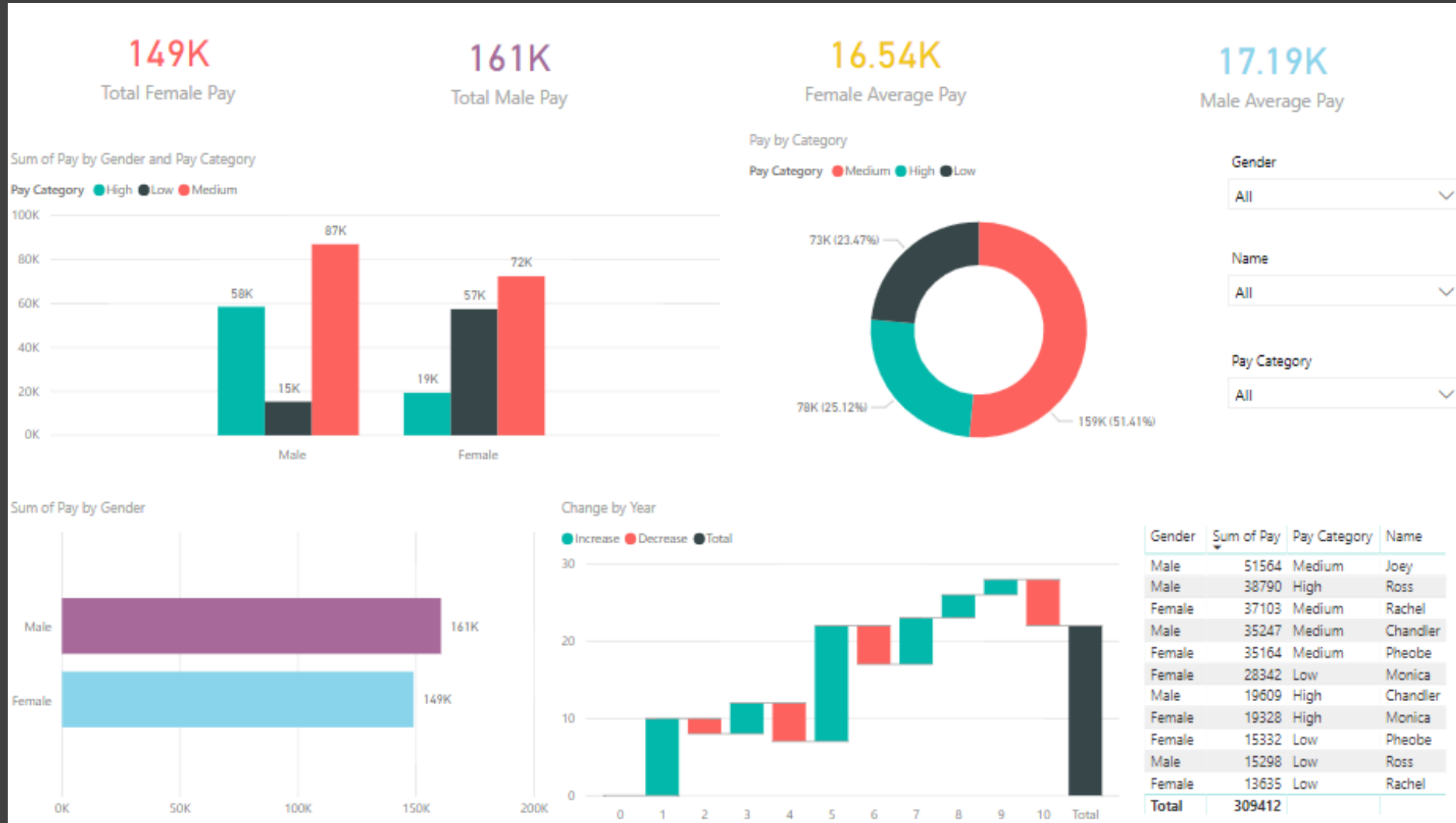


WHY POWER BI?

- MICROSOFT PRODUCTS ARE **EASILY ACCESSIBLE** AND PEOPLE ARE COMFORTABLE USING MICROSOFT PRODUCTS
- **EASY-TO-USE** AND NAVIGATE WITH A **VERY GOOD USER INTERFACE**
- THE WAY THE DATA CONCEPTS ARE APPLIED ARE **NOT TOO DIFFICULT** AND ARE **FLEXIBLE**

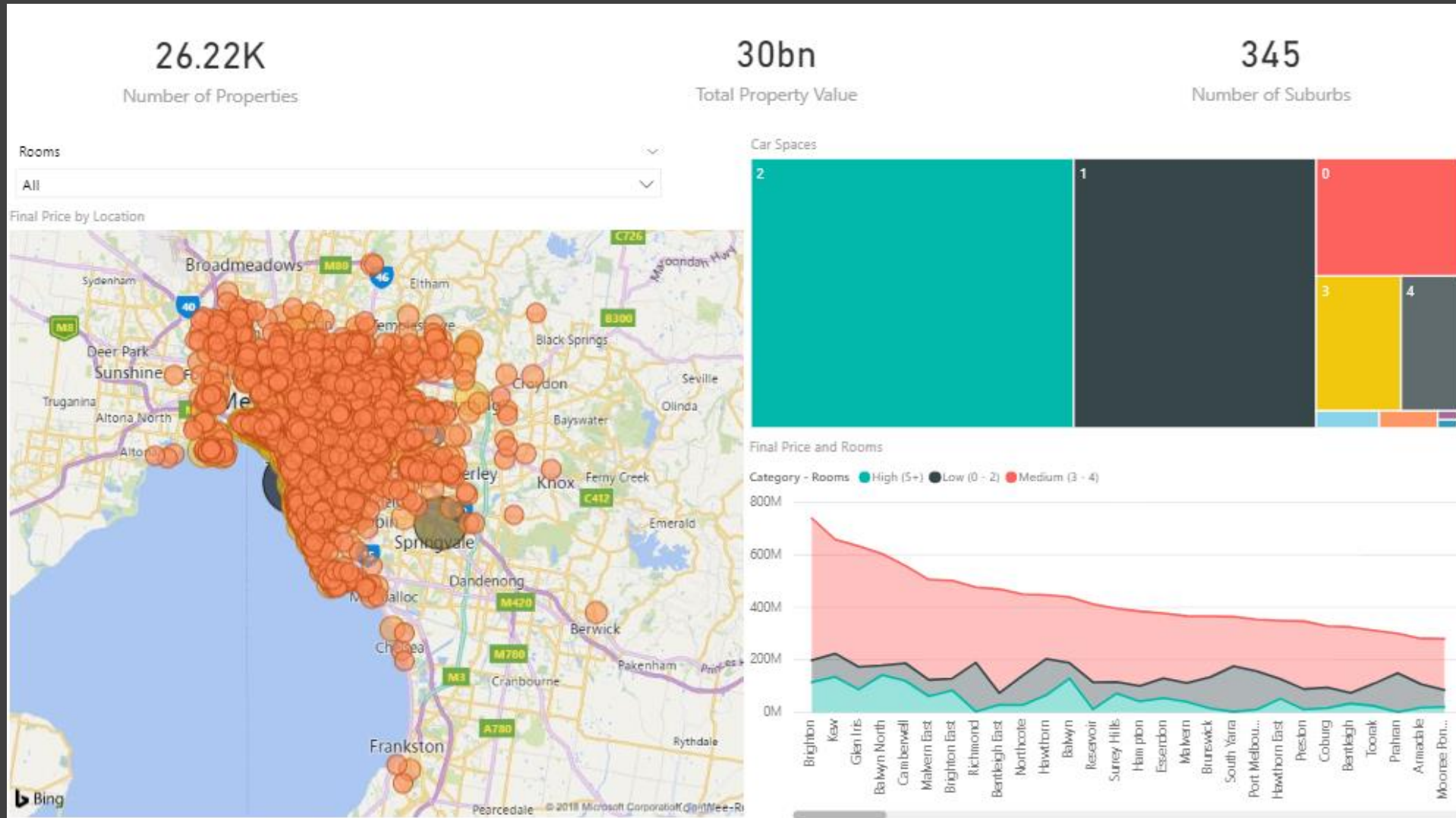


INSIGHT INTO THE FIRST DASHBOARD



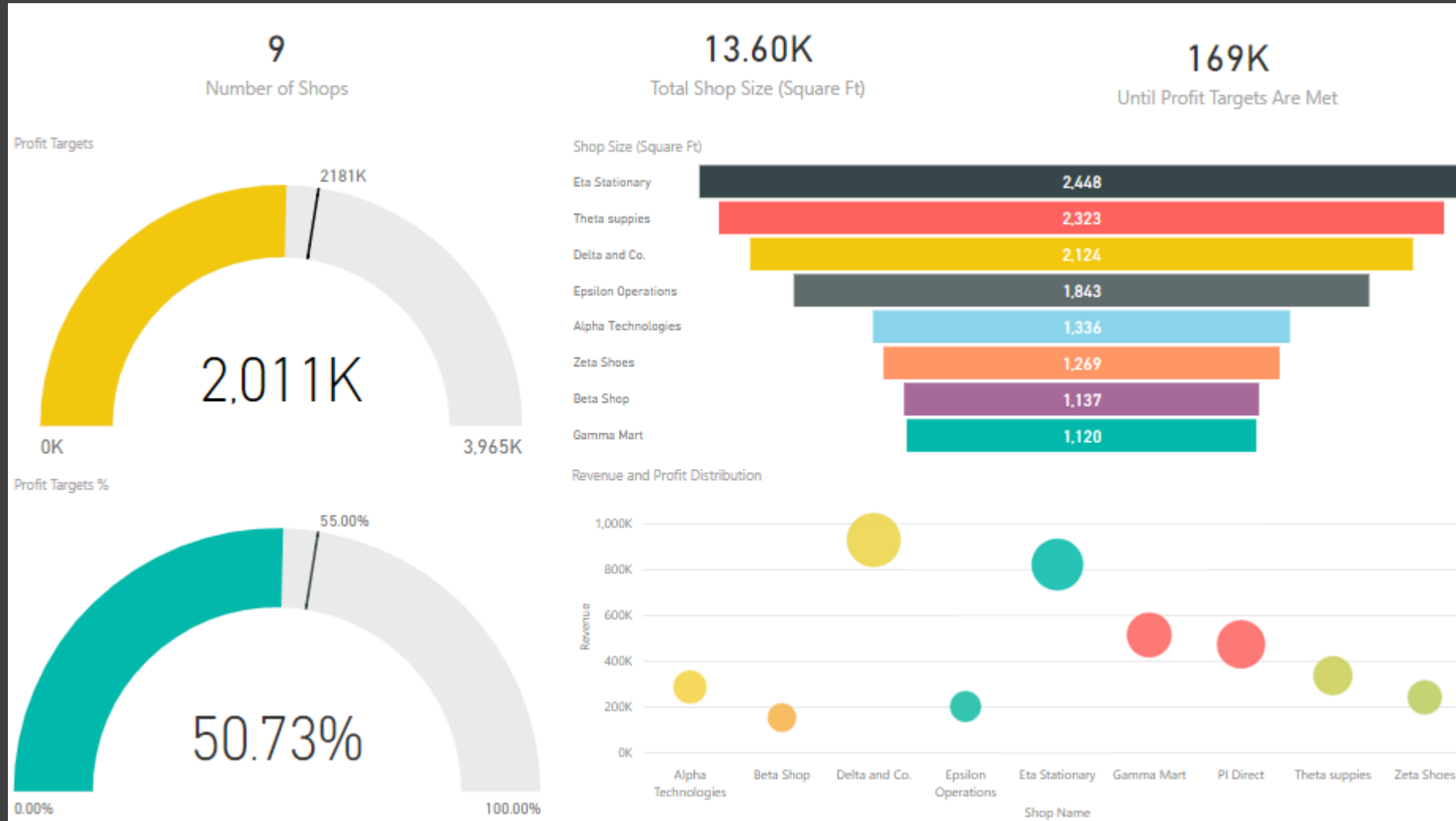


INSIGHT INTO THE SECOND DASHBOARD





INSIGHT INTO THE THIRD DASHBOARD



2. APPLYING DATA CONCEPTS



DATA CONCEPTS

- THE DATA CONCEPTS WHICH WE LEARNT IN THE SQL COURSES WILL BE APPLIED WHEN CREATING AND FINE-TUNING OUR DASHBOARDS
- THE CONCEPTS WHICH WILL PRIMARILY BE USED ARE:
 - **CONDITIONAL COLUMNS** (SIMILAR TO THE CASE STATEMENT)
 - **USING RELATIONSHIPS** (SAME CONCEPT AS JOINING TABLES)



CONDITIONAL COLUMNS

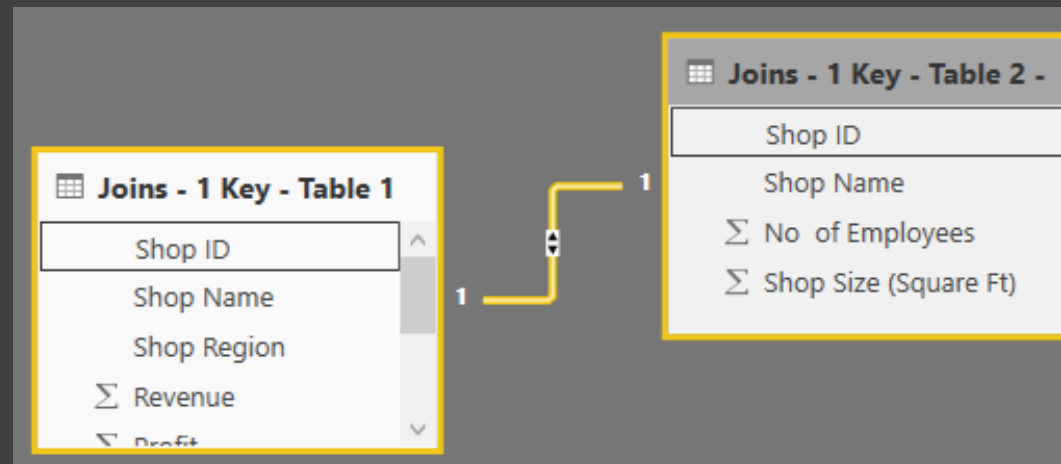
- THE 'PAY CATEGORY' COLUMN IS A **CONDITIONAL COLUMN** WHICH HAS BEEN CREATED
- THE PAY HAS BEEN CATEGORISED AS 'LOW', 'MEDIUM' AND 'HIGH'
- CATEGORISING **ADDS AN EXTRA DYNAMIC** TO OUR VISUALISATION

Name ▾	Pay ▾	Gender ▾	Pay Category ▾
Pheobe	18625	Female	Medium
Rachel	18616	Female	Medium
Rachel	18487	Female	Medium
Monica	19328	Female	High
Rachel	13635	Female	Low
Monica	14676	Female	Low
Monica	13666	Female	Low
Pheobe	16539	Female	Medium
Pheobe	15332	Female	Low
Joey	18498	Male	Medium
Joey	16533	Male	Medium
Joey	16533	Male	Medium
Ross	19728	Male	High
Ross	15298	Male	Low
Ross	19062	Male	High
Chandler	19609	Male	High
Chandler	16584	Male	Medium
Chandler	18663	Male	Medium



RELATING TABLES

- TABLES ARE LINKED TOGETHER BY '**RELATING**' THEM. THIS IS A VERY SIMILAR CONCEPT TO A JOIN, BUT A NEW JOINED TABLE IS NOT CREATED
- LIKE JOINS, **WE HAVE TO DEFINE WHICH COLUMNS ARE BEING USED TO LINK THE TABLES** WHEN THEY ARE BEING RELATED (SIMILAR TO A JOIN KEY)





RELATING TABLES

- THERE ARE DIFFERENT TYPES OF RELATIONSHIPS TABLES CAN HAVE:
 - a) **ONE-TO-ONE:** LINKING ONE MATCHING RECORD FROM THE PRIMARY TABLE TO A MATCHING RECORD IN THE SECONDARY TABLE
 - b) **MANY-TO-ONE:** LINKING MORE THAN ONE MATCHING RECORD FROM THE PRIMARY TABLE TO ONE RECORD IN THE SECONDARY TABLE
 - c) **ONE-TO-MANY:** LINKING ONE MATCHING RECORD FROM THE PRIMARY TABLE TO MORE THAN ONE MATCHING RECORD IN THE SECONDARY TABLE
 - d) **MANY-TO-MANY:** THIS RELATIONSHIP WILL NOT BE COVERED IN THIS COURSE

One to one (1:1)	▼
Many to one (*:1)	
One to one (1:1)	
One to many (1:*)	
Many to Many (*:*)	



A) ONE-TO-ONE

- A ONE-TO-ONE RELATIONSHIP IS WHERE WE HAVE **ONE RECORD FROM THE PRIMARY TABLE LINKING TO THE MATCHING RECORD FROM THE SECONDARY TABLE**
- ONLY ONE RECORD IS LINKED BETWEEN BOTH TABLES
- BOTH THE PRIMARY AND SECONDARY TABLES HAVE UNIQUE ENTRIES IN THE LINKING COLUMN

ID	Pay
1	1700
2	1800
3	1750



Employee ID	Name	Age
1	Raj	20
2	John	22
3	Tunde	21





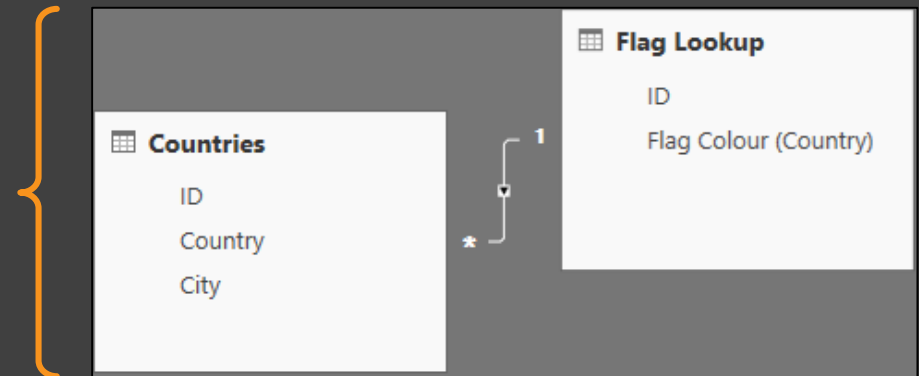
B) MANY-TO-ONE

- A MANY-TO-ONE RELATIONSHIP IS WHERE WE HAVE **MORE THAN ONE RECORD FROM THE PRIMARY TABLE LINKING TO ONE MATCHING RECORD FROM THE SECONDARY TABLE**
- THE PRIMARY TABLE DOES NOT HAVE UNIQUE ENTRIES IN THE LINKING COLUMN
- THE SECONDARY TABLE HAS UNIQUE ENTRIES IN THE LINKING COLUMN

ID	Country	City
1	England	London
1	England	Manchester
1	England	Liverpool
2	Scotland	Edinburgh
2	Scotland	Aberdeen
2	Scotland	Glasgow
3	Wales	Newport
3	Wales	Cardiff



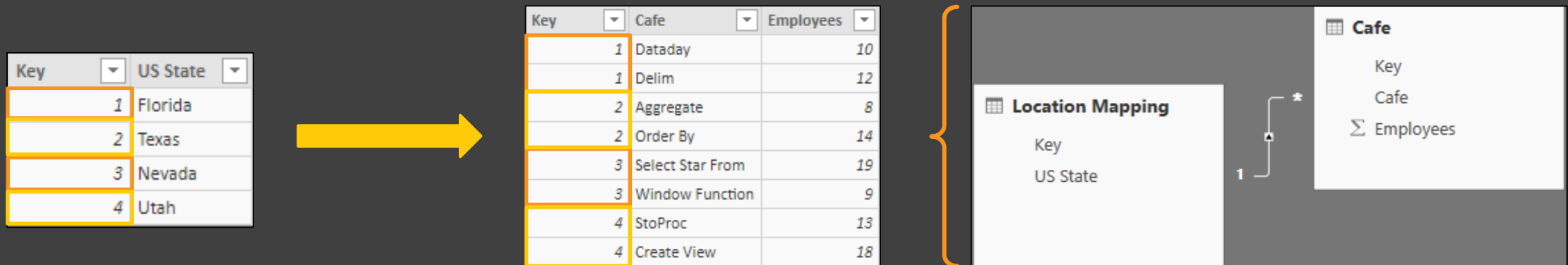
ID	Flag Colour (Country)
1	Red and White
2	Blue and White
3	Red, Green and White





C) ONE-TO-MANY

- A ONE-TO-MANY RELATIONSHIP IS WHERE WE HAVE **MORE THAN ONE RECORD FROM THE SECONDARY TABLE LINKING TO ONE MATCHING RECORD FROM THE PRIMARY TABLE**
- THE PRIMARY TABLE HAS UNIQUE ENTRIES IN THE LINKING COLUMN
- THE SECONDARY TABLE DOES NOT HAVE UNIQUE ENTRIES IN THE LINKING COLUMN

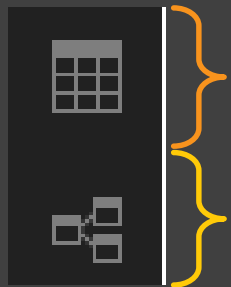


3. BACKEND OF POWER BI



BACKEND

- THE BACKEND IS WHERE WE **CARRY OUT OPERATIONS TO ENSURE THE CORRECT DATA IS BEING USED FOR OUR VISUALS**
- IT IS WHERE THE FOLLOWING TAKES PLACE:
 - IMPORTING DATA
 - MANAGING TABLE RELATIONSHIPS IN THE DATA MODEL
 - MODIFYING TABLES AND COLUMNS



DATA TAB — MANAGE DATA

RELATIONSHIPS TAB — MANAGE TABLE RELATIONSHIPS

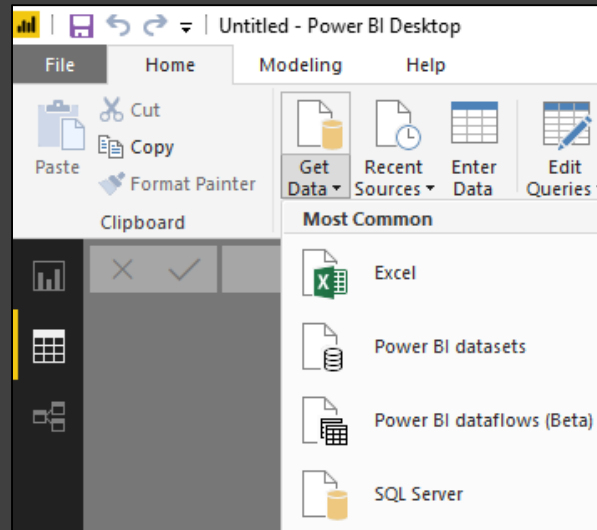
4. DATA PREPARATION PART A



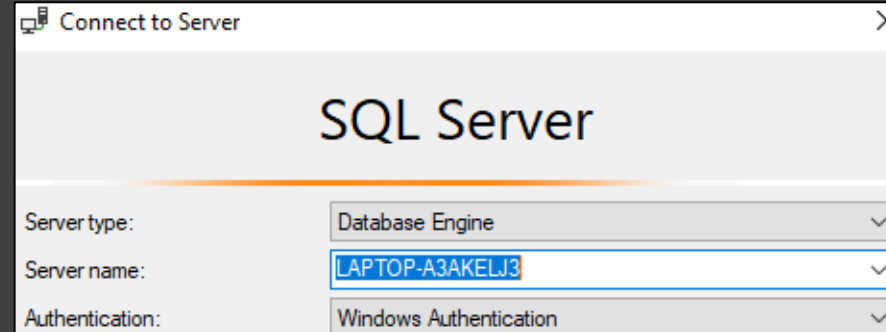
IMPORTING DATA

- IN POWER BI, DATA CAN BE IMPORTED FROM MULTIPLE SOURCES SUCH AS:
 - EXCEL FILES
 - CSV FILES
 - DIRECTLY FROM SQL DATABASES
- WE WILL BE IMPORTING **DIRECTLY FROM THE SQL DATABASE** WHICH WE HAVE WORKED WITH IN THE SQL COURSES

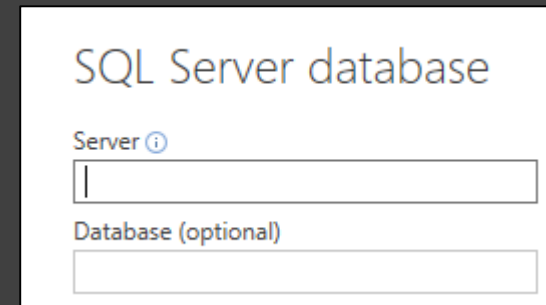
IMPORTING DATA



CLICK ON 'SQL SERVER'
AND THE FOLLOWING
WINDOW WILL APPEAR

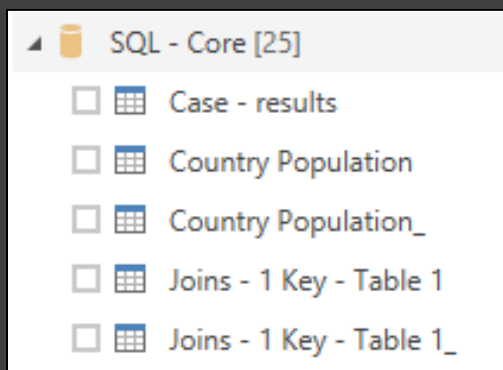


WE NEED TO PUT IN THE
NAME OF THE SERVER AS IT
APPEARS IN SQL

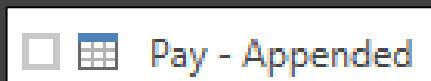




IMPORTING DATA



SELECT THE DATABASE
AND SELECT THE 'PAY -
APPENDED' TABLE



A PREVIEW OF THE
TABLE WILL APPEAR

Pay - Appended

Name	Pay
Pheobe	18625
Rachel	18616
Rachel	18487
Monica	19328
Rachel	13635
Monica	14676
Monica	13666
Pheobe	16539
Pheobe	15332
Joey	18498
Joey	16533
Joey	16533
Ross	19728
Ross	15298
Ross	19062
Chandler	19609
Chandler	16584
Chandler	18663

PRESS THE
'LOAD' BUTTON



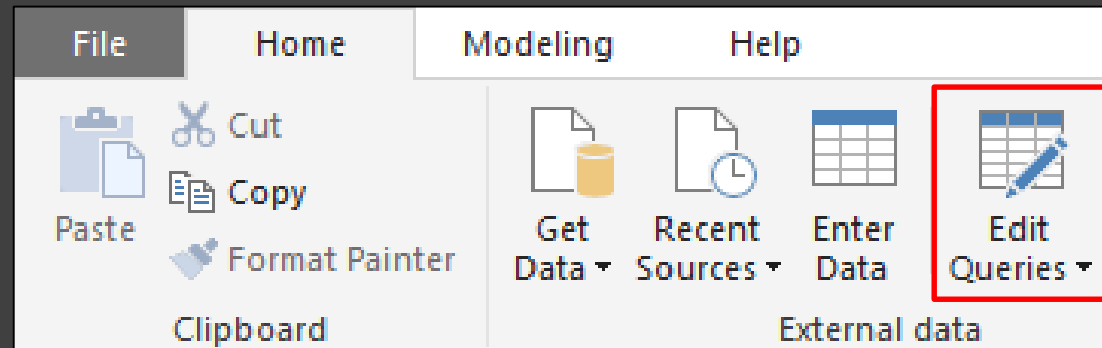
TABLE
LOADED

Name	Pay
Pheobe	18625
Rachel	18616
Rachel	18487
Monica	19328
Rachel	13635
Monica	14676



DATA OPERATIONS

- TO CARRY OUT DATA OPERATIONS SUCH AS CREATING CONDITIONAL COLUMNS AND CONVERTING DATA TYPES **WE NEED TO GO TO 'EDIT QUERIES'**





CONVERTING DATA TYPES

	A ^B _C Name	1 ² ₃ Pay
1	Pheobe	18625
2	Rachel	18616
3	Rachel	18487
4	Monica	19328



- SELECT THE COLUMN WHICH NEEDS TO HAVE ITS DATA TYPE CONVERTED
- GO TO THE 'DATA TYPE' DROPDOWN AND PICK THE CORRECT DATA TYPE
- FOR THE 'PAY - APPENDED' TABLE, NO DATA CONVERSION HAS TO BE CARRIED OUT

The screenshot shows the 'Transform' tab in the software interface. The 'Data Type' dropdown menu is open, displaying a list of data types. The 'Pay' column in the 'Pay - Appended' table is highlighted, and the 'Data Type' dropdown is set to 'Whole Number'.

File	Home	Transform	Add Column	View	Help
Close & Apply	New Source	Recent Sources	Enter Data	Data source settings	Manage Parameters
Close	New Query	Data Sources	Parameters	Refresh Preview	Advanced Editor
				Query	Manage
				Choose Columns	Remove Columns
				Manage Columns	
				Keep Rows	Remove Rows
				Reduce Rows	
				Sort	
				Split Column	Group By

Queries [1]

- Pay - Appended

A ^B _C Name	1 ² ₃ Pay
Pheobe	18625
Rachel	18616
Rachel	18487
Monica	19328
Rachel	13635
Monica	14676
Monica	13666
Pheobe	16539

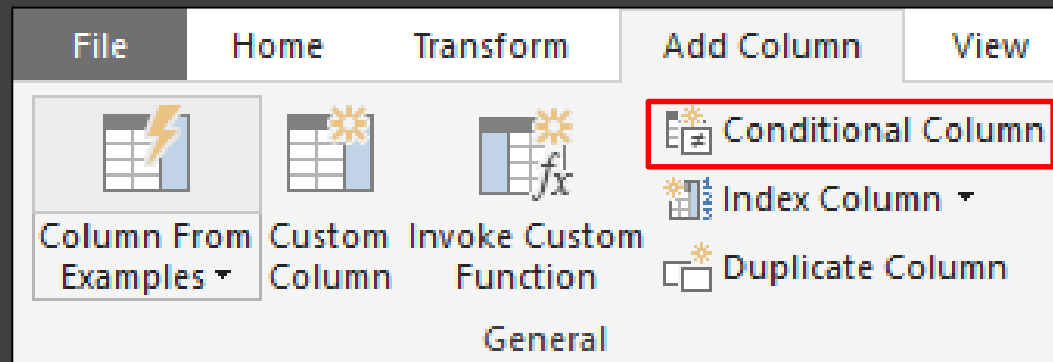
Data Type: Whole Number

- Decimal Number
- Fixed decimal number
- Whole Number
- Percentage
- Date/Time
- Date
- Time
- Date/Time/Timezone
- Duration
- Text
- True/False
- Binary



CREATING CONDITIONAL COLUMNS

- CREATING A CONDITIONAL COLUMN **ALLOWS FOR DATA TO BE CATEGORISED**. THIS ALLOWS ADDITIONAL DYNAMICS TO BE INTRODUCED TO VISUALISATIONS
- CLICK ON THE 'ADD COLUMN' SECTION AND CLICK ON 'CONDITIONAL COLUMN'





CREATING CONDITIONAL COLUMNS

- THE CONDITIONS HAVE TO BE INSERTED INTO THE AREAS OUTLINED IN RED. THE AREA OUTLINED IN ORANGE IS WHERE WE NAME THE NEW CONDITIONAL COLUMN

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Pay Category

	Column Name	Operator	Value		Output
If	Pay	is greater than or...	ABC 123 19000	Then	ABC 123 High
Else If	Pay	is less than or equ...	ABC 123 15500	Then	ABC 123 Low

Add rule

Otherwise
ABC 123 Medium

OK Cancel



CREATING CONDITIONAL COLUMNS

- THE CONDITIONAL COLUMN HAS BEEN CREATED BASED ON THE CONDITIONS WHICH WE DEFINED (LOW, MEDIUM AND HIGH)
- WE DO NOT NEED TO CARRY OUT ANY DATA CONVERSIONS
- **ALWAYS HAVE IN THE BACK OF YOUR MIND IF DATA TYPES NEED TO BE CONVERTED!**

Untitled - Power Query Editor

File Home Transform Add Column View Help

Close & Apply Close New Source Recent Sources Enter Data New Query Data source settings Data Sources Manage Parameters Parameters Refresh Preview Query Properties Advanced Editor Manage

Queries [1] Pay - Appended

	ABC Name	123 Pay	ABC 123 Pay Category
1	Pheobe	18625	Medium
2	Rachel	18616	Medium
3	Rachel	18487	Medium
4	Monica	19328	High
5	Rachel	13635	Low
6	Monica	14676	Low
7	Monica	13666	Low
8	Pheobe	16539	Medium
9	Pheobe	15332	Low
10	Joey	18498	Medium
11	Joey	16533	Medium
12	Joey	16533	Medium
13	Ross	19728	High
14	Ross	15298	Low
15	Ross	19062	High
16	Chandler	19609	High
17	Chandler	16584	Medium
18	Chandler	18663	Medium



CREATING CONDITIONAL COLUMNS

- ANOTHER CONDITIONAL COLUMN TO HIGHLIGHT THE GENDER OF THE PERSON WHO IS BEING PAID CAN BE CREATED

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

	Column Name	Operator	Value ?	Output ?
If	Name	equals	ABC 123 Monica	Then ABC 123 Female
Else If	Name	equals	ABC 123 Rachel	Then ABC 123 Female
Else If	Name	equals	ABC 123 Pheobe	Then ABC 123 Female

Add rule

Otherwise ?

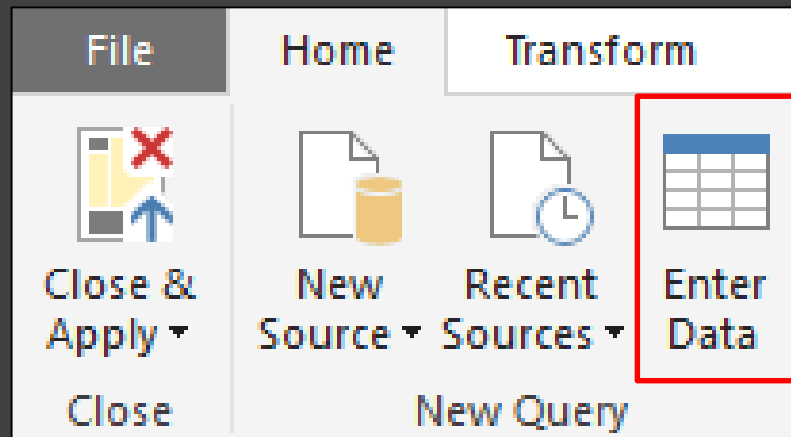
OK

Cancel



CREATING TABLES

- WE CAN **MANUALLY CREATE TABLES** IN POWER BI. THIS IS NOT VERY COMMON
- IT IS USED IF A **LOOKUP TABLE** OR A TABLE OF SIMILAR NATURE IS INVOLVED





CREATING TABLES

	Year	Change	*
1	0	0	
2	1	10	
3	2	-2	
4	3	4	
5	4	-5	
6	5	15	
7	6	-5	
8	7	6	
9	8	3	
10	9	2	
11	10	-6	
*			

Name:

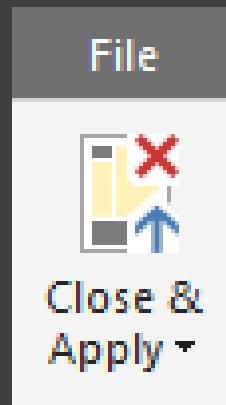
OK Cancel

- ONCE THE 'ENTER DATA' BUTTON IS CLICKED, A WINDOW WILL APPEAR PROMPTING YOU TO ENTER DATA
- ENTER THE DATA VALUES AS SHOWN
- THE ASTERISK IN THE RED BOX ALLOWS FOR MORE COLUMNS TO BE ADDED
- THE AREA HIGHLIGHTED BY THE ORANGE BOX IS WHERE THE NAME OF THE TABLE IS ENTERED



APPLYING CHANGES

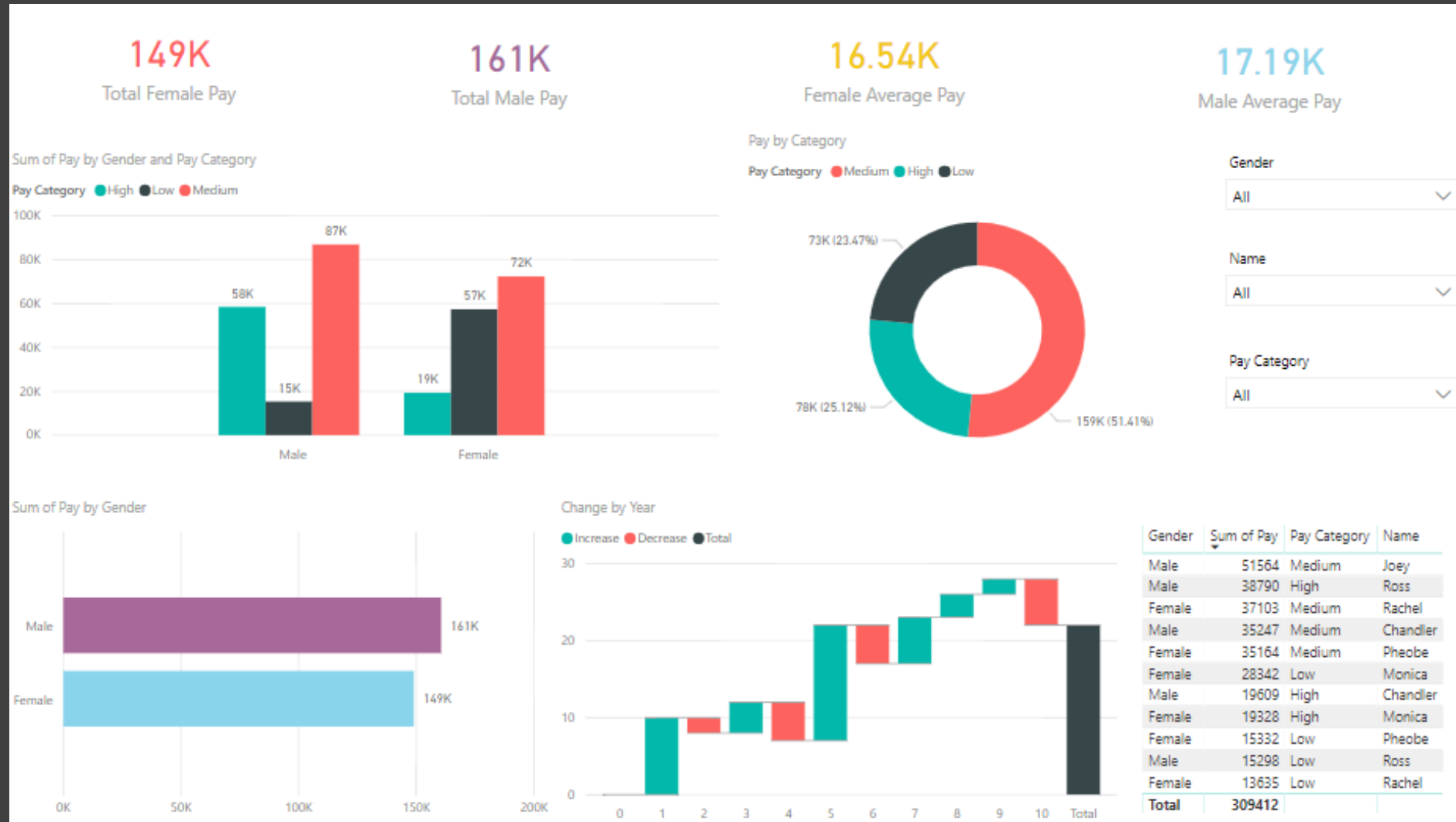
- AFTER THE DATA OPERATIONS HAVE BEEN CARRIED OUT, IT IS **ESSENTIAL** THAT THOSE CHANGES BE SAVED AND APPLIED
- TO DO THIS, THE '**CLOSE & APPLY**' BUTTON NEEDS TO BE CLICKED ON



5. THE FIRST DASHBOARD: SALARY



THE FIRST DASHBOARD: SALARY





CREATING THE FIRST DASHBOARD

- THE 'PAY – APPENDED' TABLE IS USED FOR THE FIRST DASHBOARD
- IN THE FIRST DASHBOARD, WE WILL BE LOOKING AT THE FOLLOWING VISUALISATIONS:
 - CLUSTERED COLUMN CHART
 - DONUT CHART
 - CLUSTERED BAR CHART
 - WATERFALL CHART
 - TABLE
 - CARD
 - SLICER



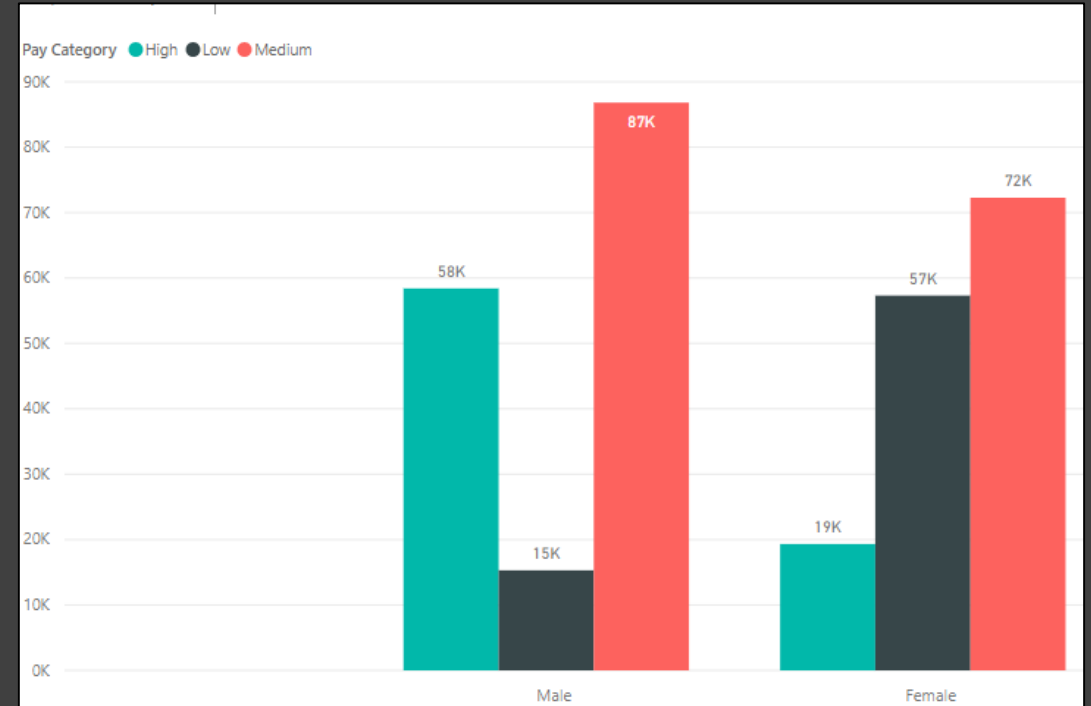
CLUSTERED COLUMN CHART

The interface shows a grid of chart types at the top. Below it, there are three main sections: 'Axis', 'Legend', and 'Value'. The 'Axis' section has a dropdown menu set to 'Gender'. The 'Legend' section has a dropdown menu set to 'Pay Category'. The 'Value' section has a dropdown menu set to 'Sum of Pay'. There is also a 'Tooltips' section with a button 'Add data fields here'. At the bottom, there is a 'FILTERS' section with a button 'Visual level filters'.

- THE CLUSTERED CHART CAN BE THOUGHT OF AS A BAR CHART
- WE CAN PRODUCE A SIMPLE BAR CHART BY USING THE 'AXIS' AND 'VALUE' SECTIONS
- USING THE 'LEGEND' SECTION INTRODUCES A NEW DYNAMIC

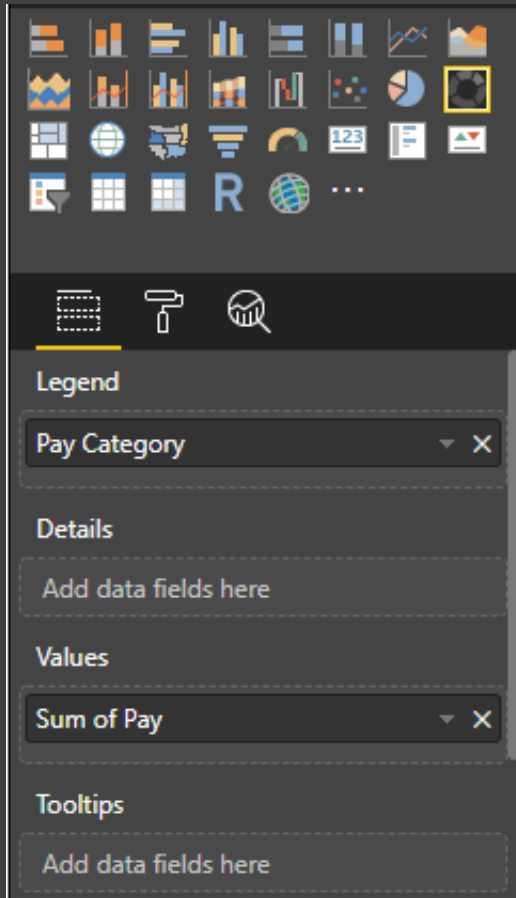


- GENDER IS ON THE X-AXIS
- SUM OF PAY IS ON THE Y-AXIS
- THE 'LEGEND' IS BASED ON THE PAY CATEGORY. THIS SPLITS THE DATA INTO THE RESPECTIVE CATEGORIES





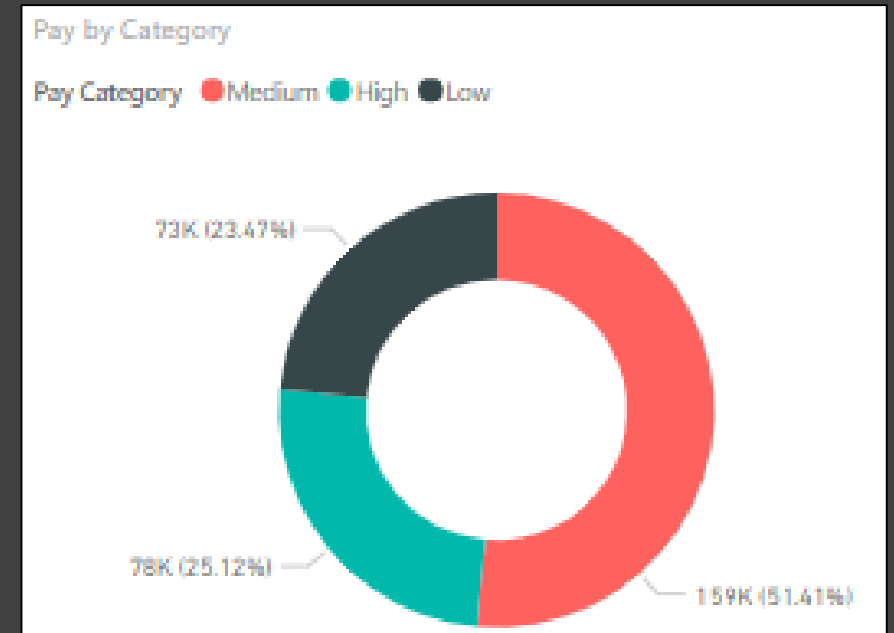
DONUT CHART



- THE DONUT CHART CAN BE THOUGHT OF AS A PIE CHART WITH A HOLE IN THE MIDDLE
- THE 'LEGEND' IS WHAT THE RESULTS ARE SPLIT BY
- THE 'VALUES' IS THE METRIC WHICH THE CHART SHOWS

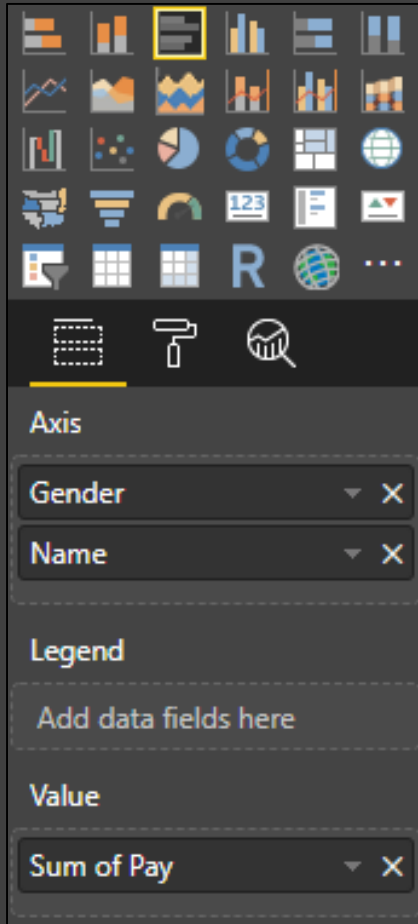


- THE DATA IS SPLIT BY THE PAY CATEGORY
- THE METRIC IS THE SUM OF PAY
- THE 'DETAILS' IS NOT ASSIGNED AS IT CAN OVERCOMPLICATE THE VISUALISATION





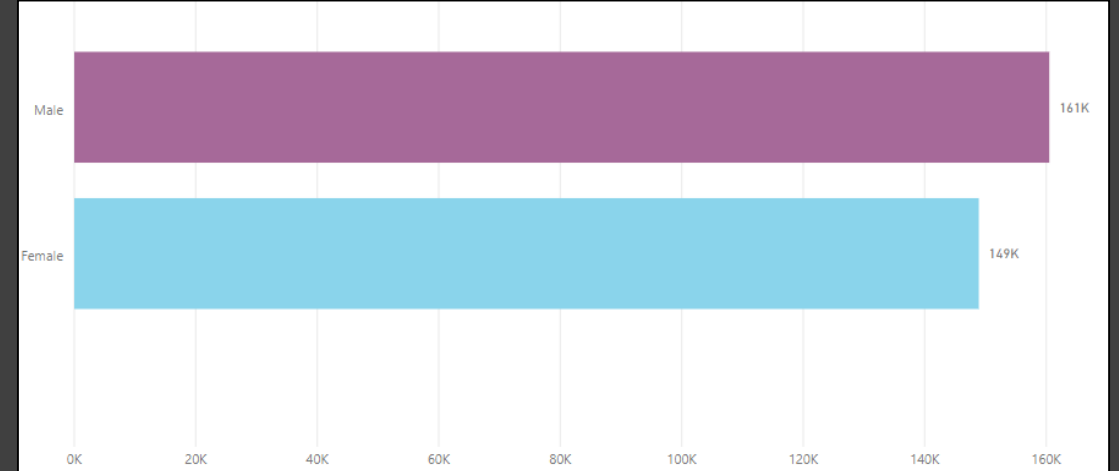
CLUSTERED BAR CHART



- THE CLUSTERED BAR CHART IS A SIDEWAYS BAR CHART
- WE ARE INTRODUCING HIERARCHY AS THERE ARE TWO COLUMNS IN THE 'AXIS' SECTION

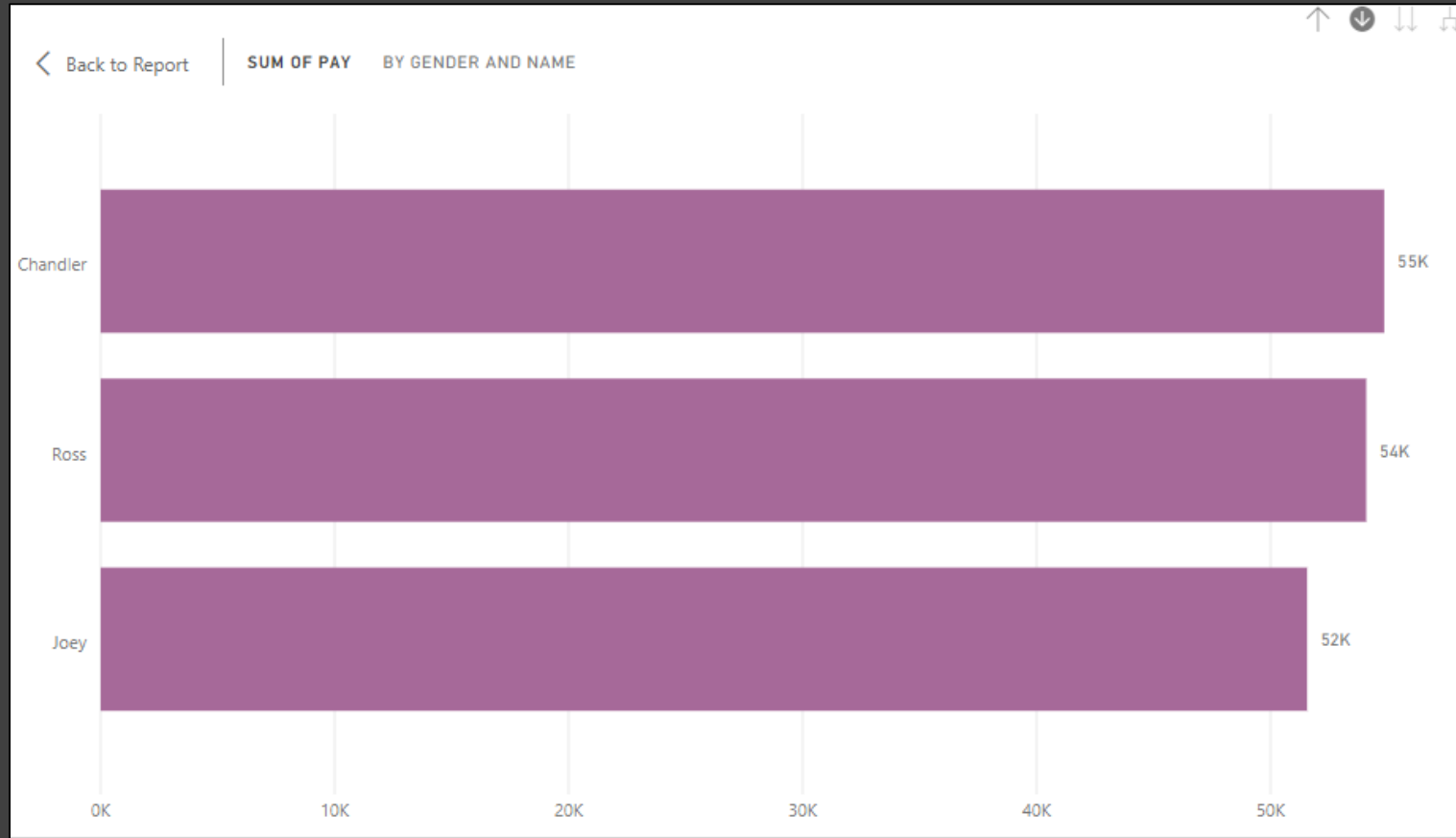


- THE SUM OF PAY IS THE 'VALUE'
- THE GENDER IS THE FIRST LAYER IN THE HIERARCHY, WHILST THE NAME IS THE SECOND LAYER IN THE HIERARCHY





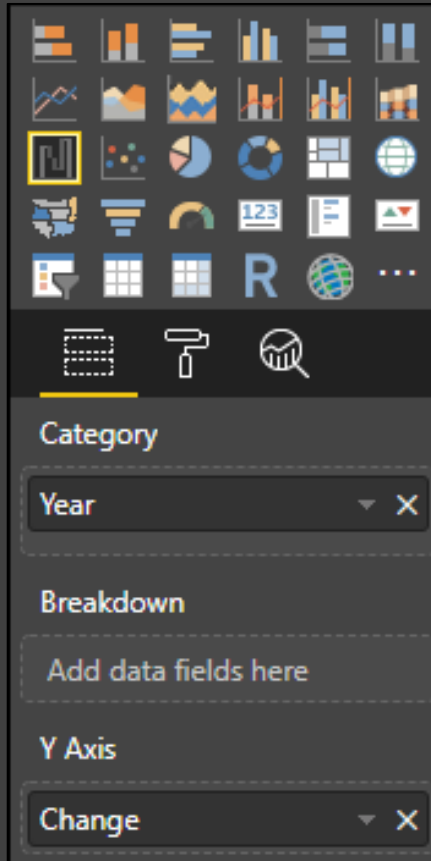
HIERARCHY (DRILLING DOWN)



- TO ENABLE THE HIERARCHAL DRILL DOWN SECTION, THE TOP RIGHT OF THE VISUALISATION HAS TO BE CONFIGURED
- CLICK THE WHITE ARROW IN THE BLACK CIRCLE TO ENABLE
- CLICKING ON THE MALE BAR WILL SHOW YOU THE THREE MALES WHICH CONTRIBUTE TO THE TOTAL



WATERFALL CHART



- THE WATERFALL CHART ALLOWS US TO SEE HOW A METRIC IS CHANGING THROUGH THE DATASET
- THE 'CATEGORY' AND 'Y AXIS' ARE DEFINED



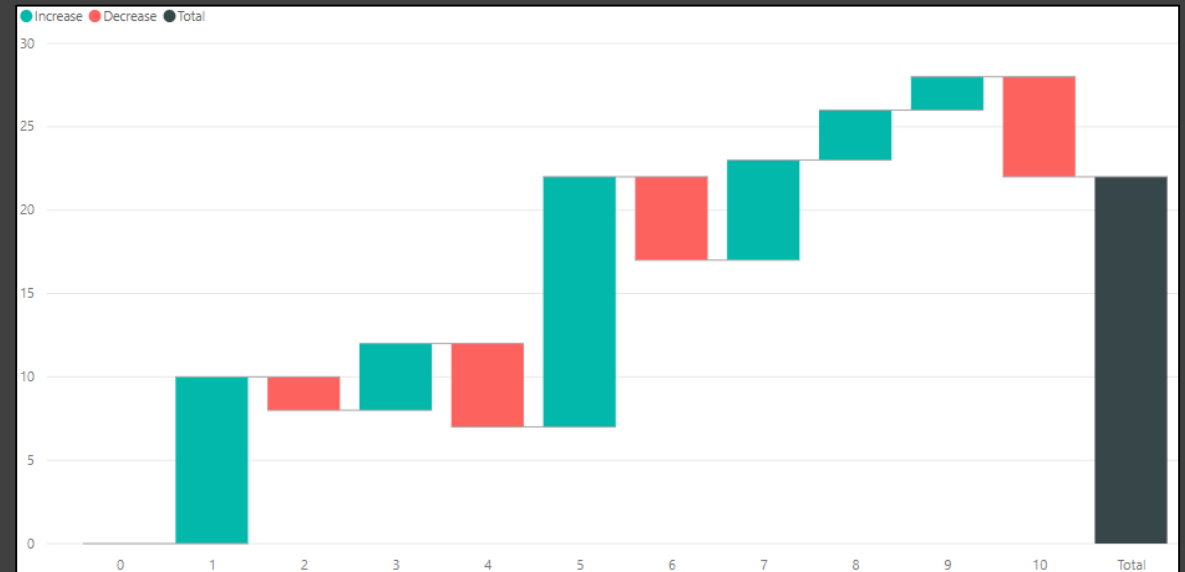
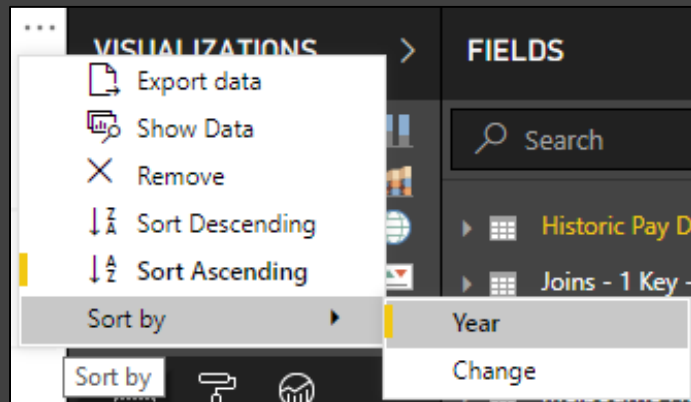
- THE GRAPH SHOWS THE DECREASE, INCREASE AND FINALLY THE TOTAL
- THIS ISN'T THE BEST WAY TO SHOW THE DATA AND IT NEEDS TO BE SORTED CORRECTLY



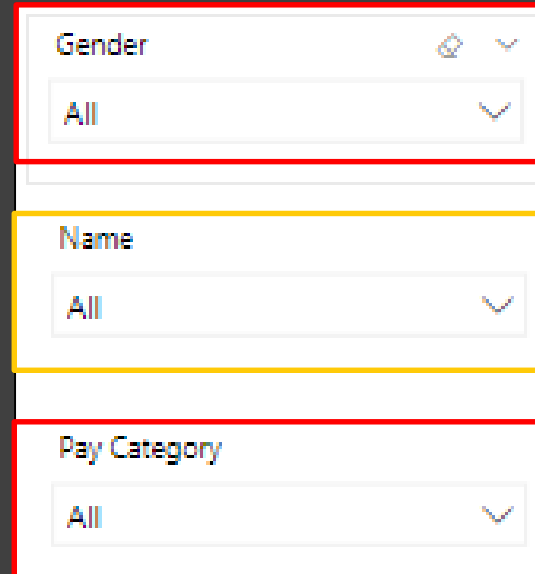
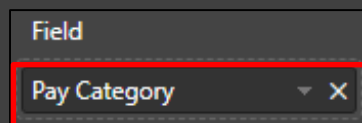
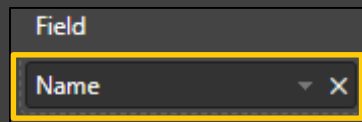
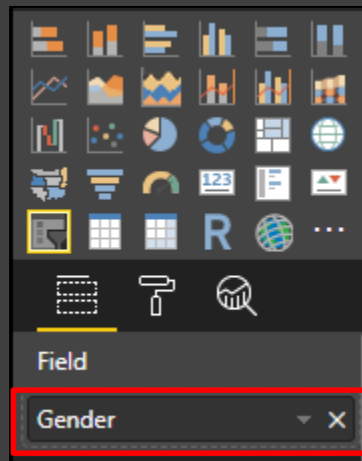


SORTING DATA

- TO SORT THE DATA, CLICK ON THE WATERFALL CHART AND WE WILL SEE THREE DOTS ON THE TOP RIGHT HAND SIDE
- CLICK ON THE THREE DOTS AND A DROPDOWN MENU WILL APPEAR
- GO TO 'SORT BY' AND CHANGE IT OPTION TO YEAR
- THE CHART WILL NOW SHOW HOW THE CHANGE OCCURRED ON A YEARLY BASIS WITH THE TOTAL AT THE END



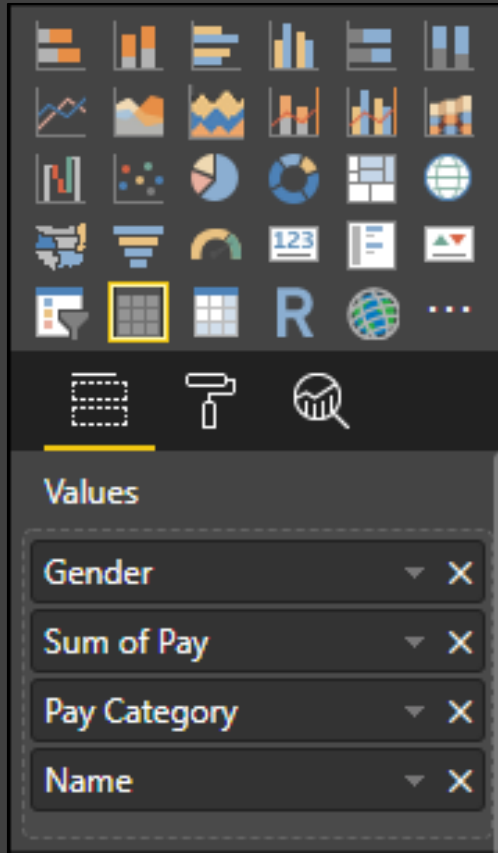
SLICER



- THE SLICER TOOL ALLOWS THE DATA TO BE FILTERED
- THE FIELD WHICH WE WANT TO BE USED FOR FILTERING HAS TO BE DRAGGED INTO THE 'FIELD' SECTION
- AS AN EXAMPLE, IF WE WANTED THE DATA TO BE FILTERED TO SEE THE PAY METRICS FOR MALES, THEN WE CAN DO SO BY CONFIGURING THE GENDER FILTER



TABLE



- THE DATA CAN BE DISPLAYED IN A TABLE
- THIS IS GOOD BECAUSE WE CAN SEE HOW THE DATA CHANGES VISUALLY AND ALSO IN THE TABLE

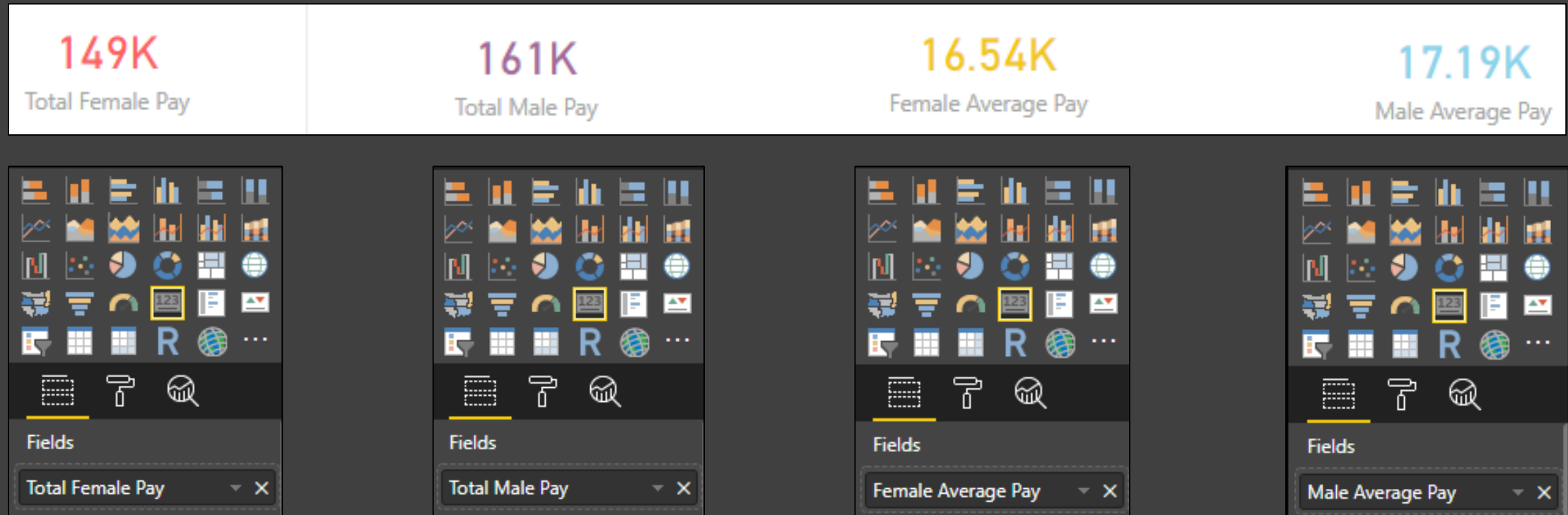
- THE COLUMNS WHICH WE WANT TO SEE HAVE TO BE DRAGGED INTO THE 'VALUES' SECTION
- THE ORDER IN WHICH WE DRAG THE COLUMNS WILL BE REFLECTED IN THE TABLE

Gender	Sum of Pay	Pay Category	Name
Male	51564	Medium	Joey
Male	38790	High	Ross
Female	37103	Medium	Rachel
Male	35247	Medium	Chandler
Female	35164	Medium	Pheobe
Female	28342	Low	Monica
Male	19609	High	Chandler
Female	19328	High	Monica
Female	15332	Low	Pheobe
Male	15298	Low	Ross
Female	13635	Low	Rachel
Total	309412		



CARD

- THE CARD ALLOWS HEADERS WITH RESULTS TO BE CREATED
- THIS IS VERY EFFECTIVE AS IT HIGHLIGHTS SPECIFIC METRICS STRAIGHT AWAY
- WE HAVE TO DRAG THE COLUMN INTO THE 'FIELDS' SECTION AND DEFINE IF WE WANT TO SEE THE AVERAGE, TOTAL OR ANOTHER MATHEMATICAL FUNCTION APPLIED TO THE SELECTED FIELD



6. DATA PREPARATION PART B



IMPORTING DATA

- THE 'MELBOURNE HOUSING MASTER' TABLE IS IMPORTED INTO POWER BI
- REMEMBER THAT THIS TABLE IS BEING IMPORTED DIRECTLY FROM THE SQL DATABASE
- DATA OPERATIONS SUCH AS CREATING CONDITIONAL COLUMNS, DUPLICATING COLUMNS AND DATA CONVERSIONS ARE CARRIED OUT ON THE TABLE



FILTERING

- ONCE THE TABLE HAS BEEN IMPORTED, FILTER OUT THE BLANK VALUES FOR THE LATITUDE AND LONGITUDE COLUMNS

A _C ^B Latitude	A _C ^B Longitude
-37.83581	145.19802
-37.835	145.21086
-37.84116	145.20236
-37.83737	145.18819
-37.833	145.19526
-37.84478	145.19817
-37.83113	145.19515
-37.8449	145.18933
-37.84299	145.20459



Text Filters

Search

☐ (Select All)

☐ (blank)

☒ -37.56692

☒ -37.57287

☒ -37.57555



A _C ^B Latitude	A _C ^B Longitude
-37.83581	145.19802
-37.835	145.21086
-37.84116	145.20236
-37.83737	145.18819
-37.833	145.19526
-37.84478	145.19817
-37.83113	145.19515
-37.8449	145.18933
-37.84299	145.20459
-37.83523	145.18497



CONVERTING DATA TYPES

- THE ROOMS COLUMN HAS TO HAVE ITS DATA CONVERTED FROM STRING TO NUMERIC

File	Home	Transform	Add Column	View	Help
Close & Apply Close	New Source New Query	Recent Sources	Enter Data	Data source settings Data Sources	Manage Parameters Parameters
				Refresh Preview Query	Properties Advanced Editor Manage
				Choose Columns Manage Columns	Remove Columns
				Keep Rows Reduce Rows	Remove Rows
				Sort	Split Column Group By

Queries [5]	Suburb	Address	Rooms	Type	Price	Suburb Price Average
Pay - Appended	1 Vermont	2/660 Canterbury Rd	3	t	820000	10
Historic Pay Data	2 Vermont	25 Caldwell Rd	3	h	865000	10
Joins - 1 Key - Table 1	3 Vermont	4 Barton Ct	4	h	0	10
Joins - 1 Key - Table 2 - Left	4 Vermont	5 Judy Ct	4	h	1133000	10
	5 Vermont	529 Mitcham Rd	5	h	0	10

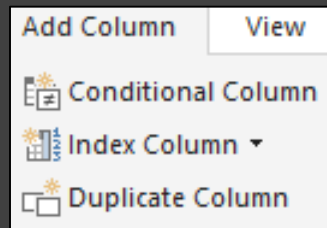


1 ² ₃ Rooms
3
3
4
4
5
4
2



CONDITIONAL COLUMN

- A CONDITIONAL COLUMN IS ADDED TO CATEGORISE THE NUMBER OF ROOMS IN A PROPERTY



Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Category - Rooms

	Column Name	Operator	Value		Output
If	Rooms	is less than	3	Then	Low (0 - 2)
Else If	Rooms	is greater than	4	Then	High (5+)

[Add rule](#)

Otherwise
Medium (3 - 4)

[OK](#) [Cancel](#)

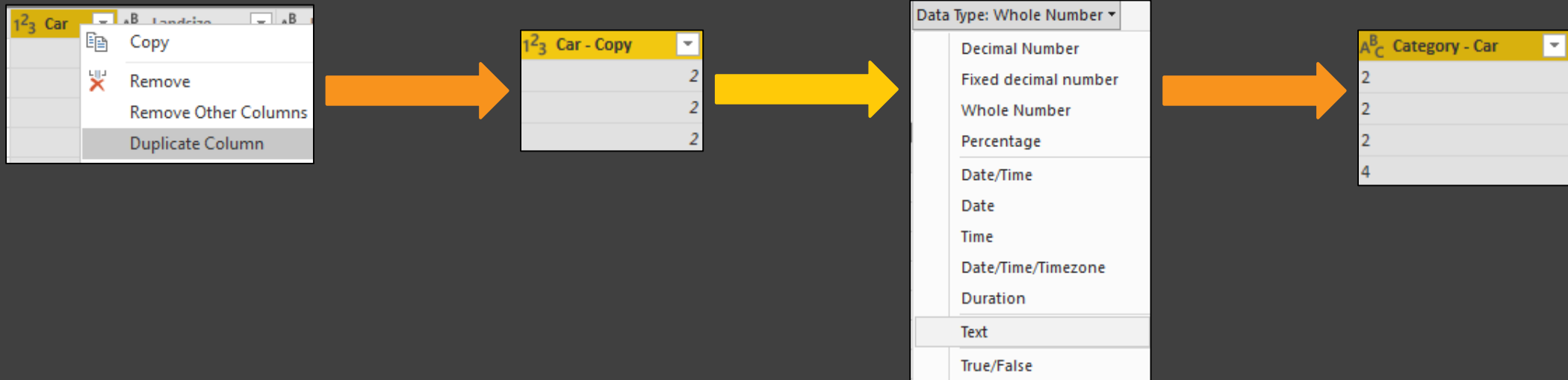


ABC 123	Category - Rooms
	Medium (3 - 4)
	Medium (3 - 4)
	Medium (3 - 4)
	Medium (3 - 4)
	High (5+)
	Medium (3 - 4)



DUPLICATING COLUMN

- THE CAR COLUMN IS DUPLICATED AND RENAMED TO 'CATEGORY – CAR'
- THE NEW COLUMN HAS ITS DATA TYPE CONVERTED FROM A NUMERIC FIELD TO A STRING FIELD





DATA
CIVILISATION

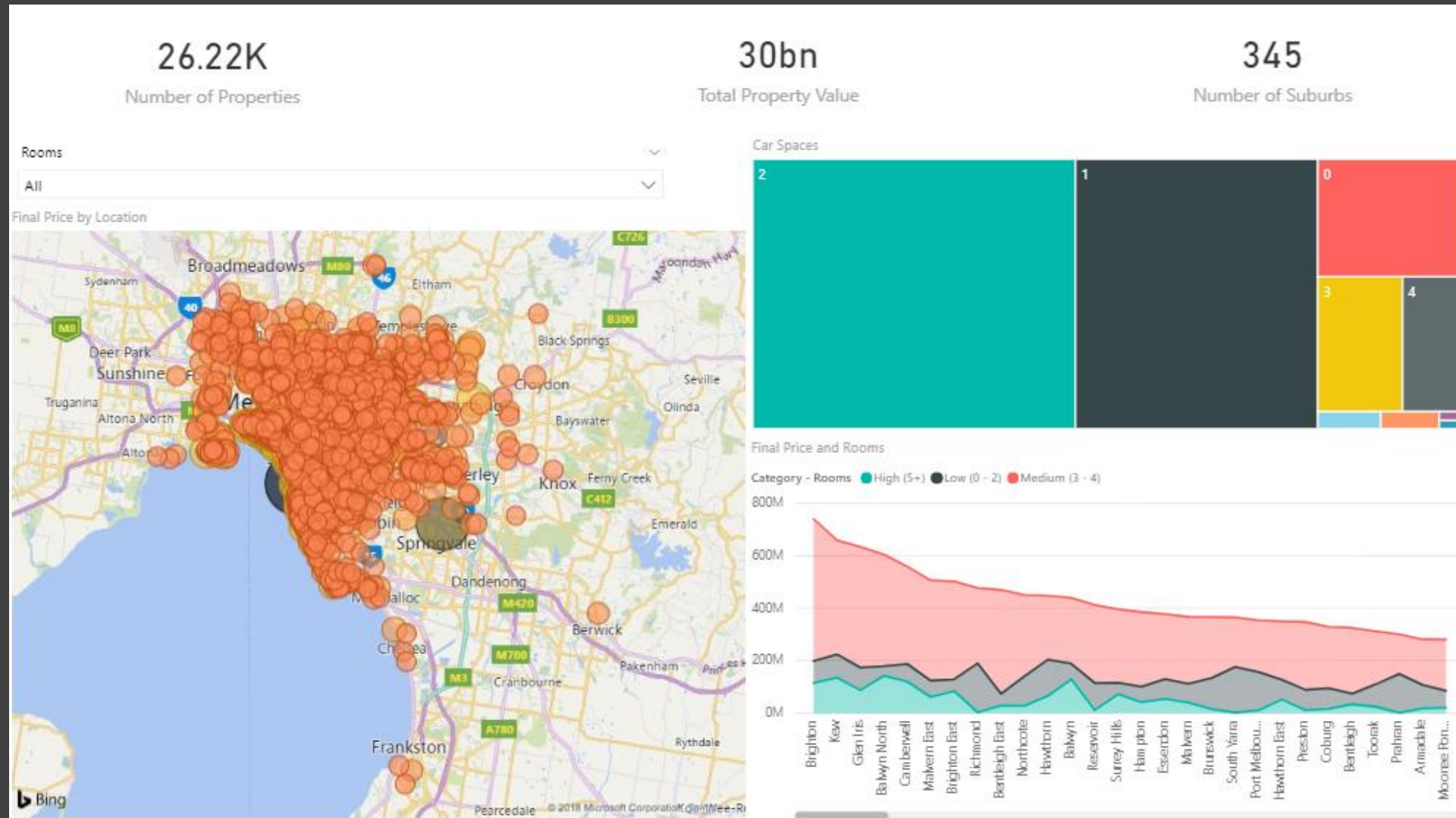
7. THE SECOND DASHBOARD: PROPERTY MARKET



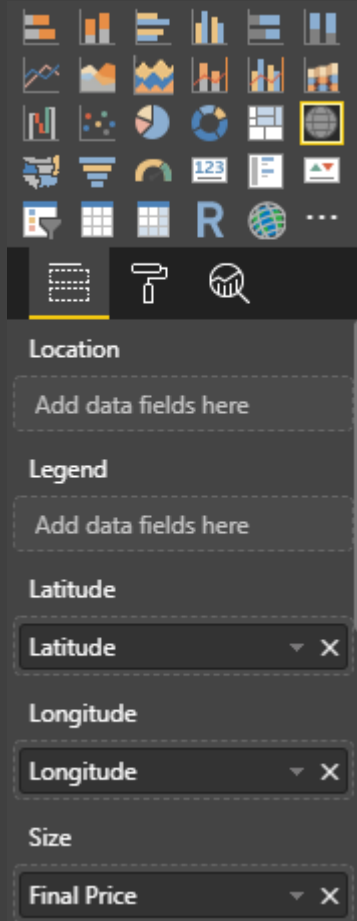
CREATING THE SECOND DASHBOARD

- THE 'MELBOURNE HOUSING MASTER' TABLE IS USED FOR THE SECOND DASHBOARD
- IN THE SECOND DASHBOARD, WE WILL BE LOOKING AT THE FOLLOWING VISUALISATIONS
 - MAP
 - TREE MAP
 - STACKED AREA CHART

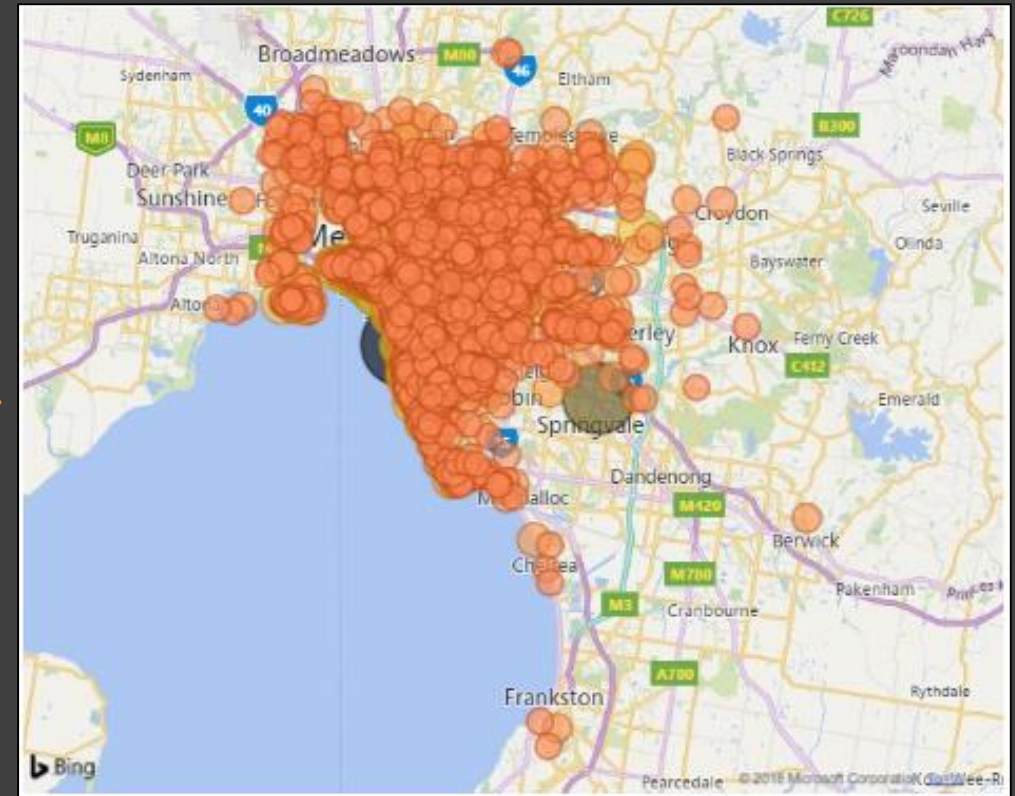
THE SECOND DASHBOARD: PROPERTY MARKET



MAP



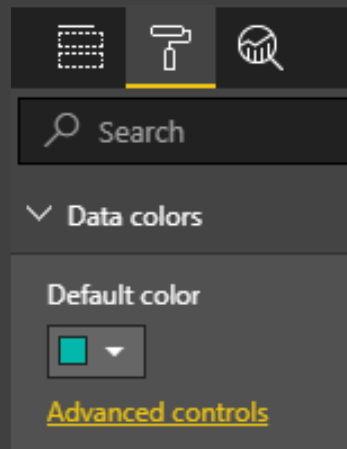
- THE MAP VISUALISATION ALLOWS METRICS TO BE SEEN BASED ON GEOGRAPHICAL LOCATION
 - THE LATITUDE AND LONGITUDE DATA IS REQUIRED TO ENSURE THIS VISUALISATION IS ACCURATE
- 
- THE LATITUDE AND LONGITUDE FIELDS FROM THE TABLES ARE USED
 - THE SIZE OF THE BUBBLE FOR EACH GEOGRAPHICAL LOCATION IS DICTATED BY THE FINAL PRICE





MAP – BUBBLE COLOURS

- TO CHANGE THE COLOURS OF THE BUBBLES, THE ADVANCED CONTROLS HAVE TO BE CONFIGURED IN THE 'DATA COLOURS' SECTION
- THE COLOUR SATURATION OF THE BUBBLES CAN BE CONFIGURED BASED ON THE VALUES IN THE DESIRED FIELD



Default color - *Data colors*

Format by Color scale ▼ [Learn more](#)

Based on field Sum of Final Price ▼

Summarization Sum ▼

Default formatting ⓘ As zero ▼

Minimum Lowest value ▼ Red ▼

Center Middle value ▼ Yellow ▼

Maximum Highest value ▼ Black ▼

☒ Diverging

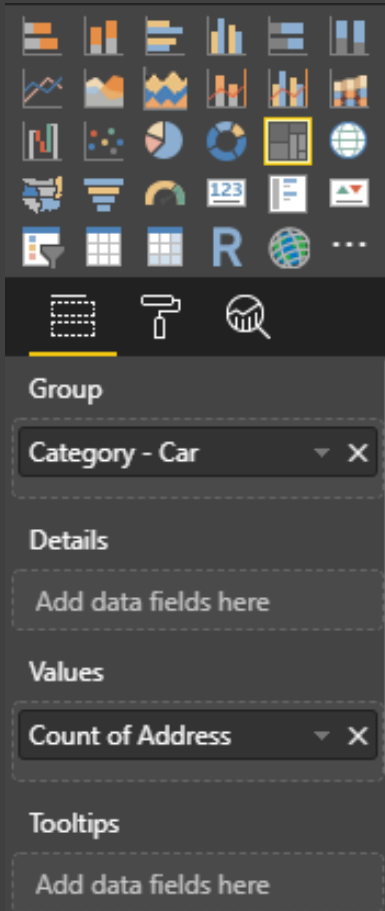
(Lowest value)

(Middle value)

(Highest value)



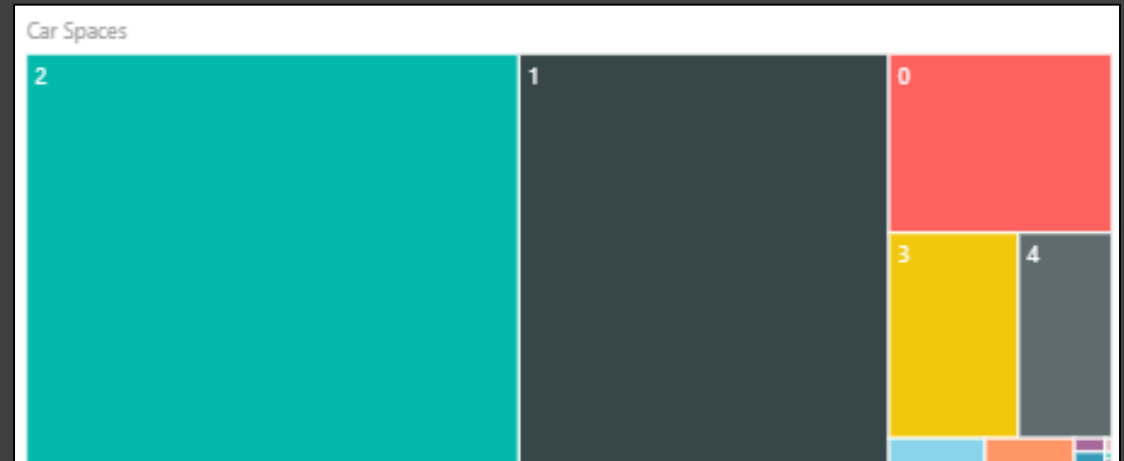
TREE MAP



- THE TREE MAP SHOWS A DISTRIBUTION OF RESULTS FOR EACH CATEGORY
- THE 'GROUP' IS WHERE THE CATEGORY IS DEFINED
- THE 'VALUES' SECTION DICTATES THE SIZE OF EACH BOX



- THE COUNT OF ADDRESSES DICTATES THE SIZE OF THE BOXES
- THE BOXES ARE SPLIT BY THE ENTRIES IN THE CATEGORY – CARS FIELD

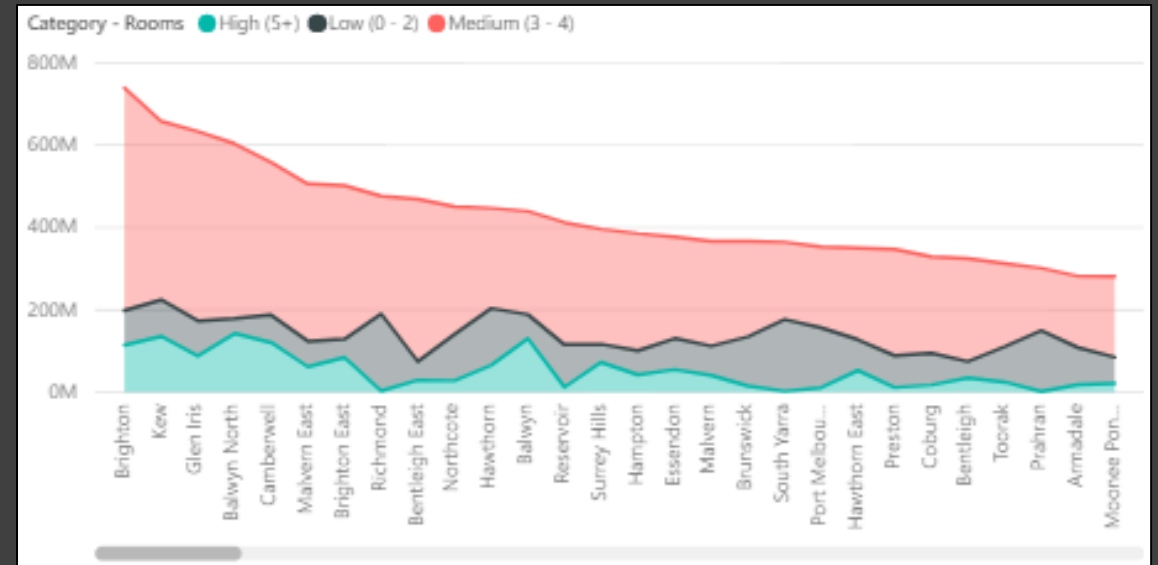




STACKED AREA CHART



- THE STACKED AREA CHART ALLOWS US TO SEE THE PROGRESSION OF A METRIC
 - THE 'AXIS', 'LEGEND' AND 'VALUE' SECTIONS ARE UTILISED
-
- THE FINAL PRICE IS ANALYSED AGAINST THE SUBURB
 - THE DOMINANCE OF THE PRICE BASED ON CATEGORY – ROOMS IS SHOWN



8. DATA PREPARATION PART C



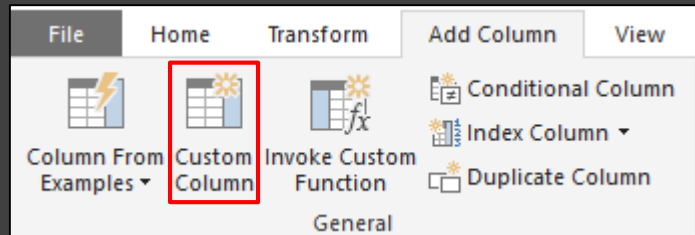
BACKGROUND

- THE 'JOINS - 1 KEY - TABLE 1' AND 'JOINS - 1 KEY - TABLE 2 - LEFT' TABLES ARE IMPORTED FROM SQL
- THE TWO TABLES HAVE A RELATIONSHIP PRESENT BETWEEN THEM
- THE 'PROFIT TARGETS' COLUMN IS CREATED TO ADD A NEW DYNAMIC TO THE DASHBOARD



CREATING CUSTOM COLUMN

- THE CUSTOM COLUMN OPTION ALLOWS US TO CREATE A COLUMN BASED ON A CALCULATION
- THE CALCULATION SHOWS THAT THE REVENUE COLUMN IS MULTIPLIED BY 0.55



- THIS NEW COLUMN SHOWS THAT THE 'PROFIT TARGETS' IS DEFINED BY BEING 55% OF THE REVENUE

Custom Column

New column name
Profit Targets

Custom column formula:
$$= ([Revenue]*55)/100$$

Available columns:
Shop ID
Shop Name
Shop Region
Revenue
Profit

<< Insert

[Learn about Power BI Desktop formulas](#)

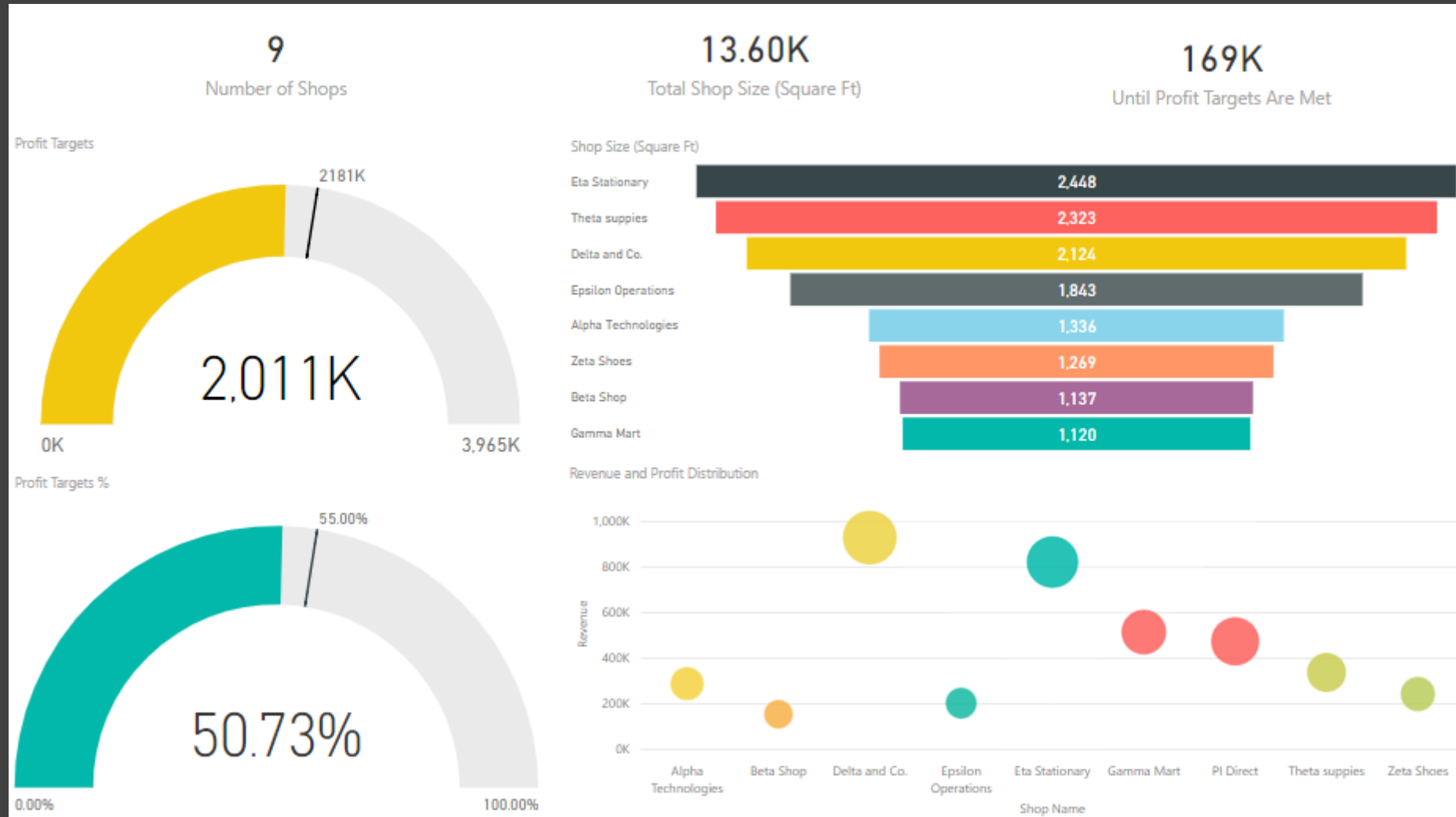
✓ No syntax errors have been detected.

OK Cancel

9. THE THIRD DASHBOARD: REVENUE



THE THIRD DASHBOARD: REVENUE



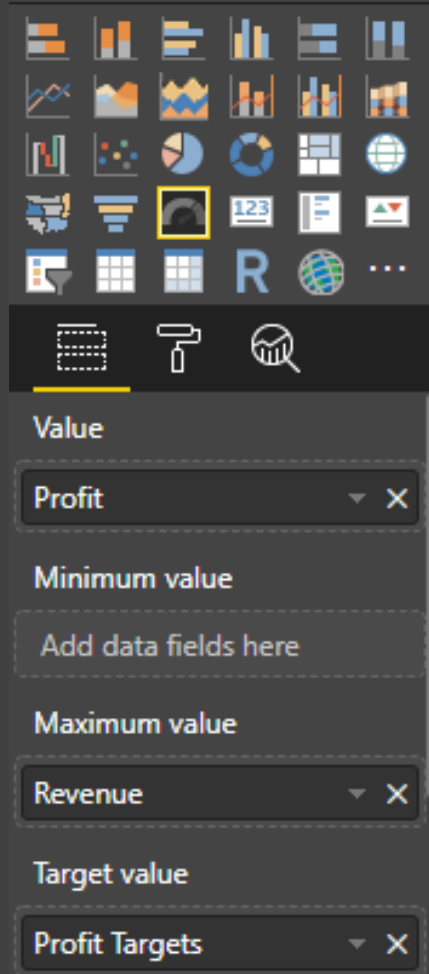


THIRD DASHBOARD

- THE 'JOINS - 1 KEY - TABLE 1' AND 'JOINS - 1 KEY - TABLE 2 - LEFT' TABLES ARE USED FOR THE THIRD DASHBOARD
- IN THE THIRD DASHBOARD, WE WILL BE LOOKING AT THE FOLLOWING VISUALISATIONS
 - GAUGE
 - FUNNEL CHART
 - SCATTER CHART

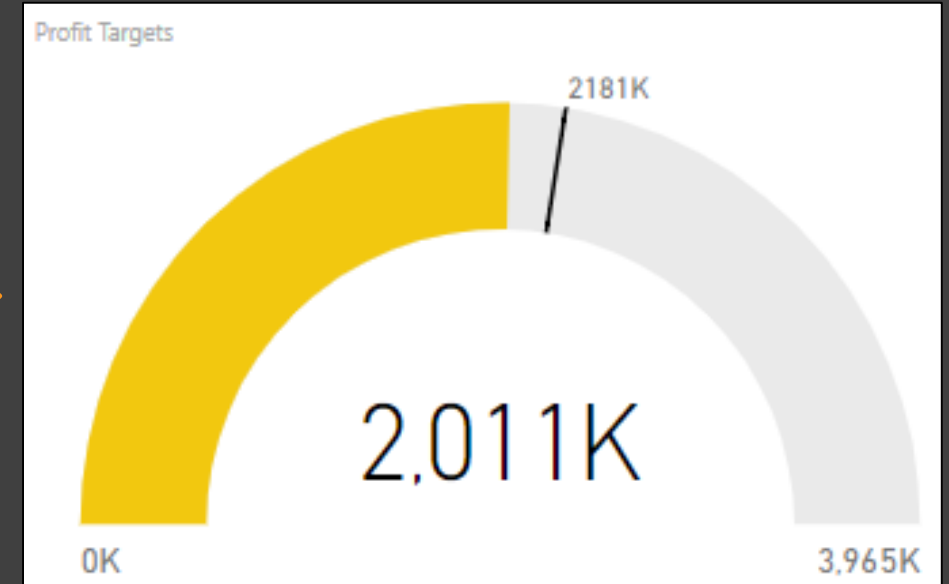


GAUGE (VALUE)



- THE GAUGE CHART SHOWS US HOW A SPECIFIC METRIC IS PROGRESSING
- THE 'VALUE' SECTION (**PROFIT**) LOOKS AT THE METRIC OF INTEREST
- THE 'MAXIMUM VALUE' SECTION (**REVENUE**) LOOKS AT THE VALUE WE ARE COMPARING THE 'VALUE' METRIC TO

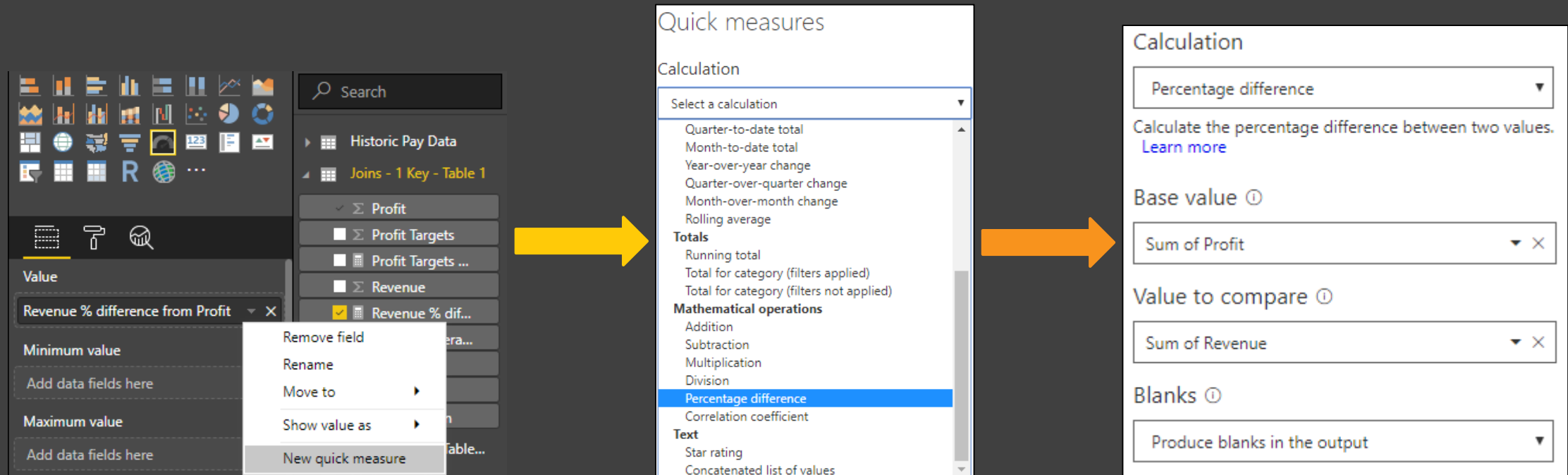
- THE 'TARGET VALUE' (PROFIT TARGETS) ADDS A SECOND LAYER OF ANALYSIS IF WE WANT TO SET A TARGET TO SEE HOW CLOSE WE ARE TO THIS TARGET





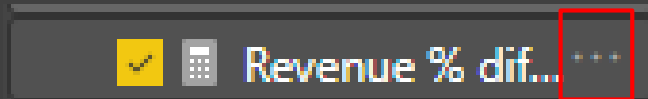
GAUGE (PERCENTAGE)

- THE GAUGE CAN ALSO SHOW PERCENTAGES TO SHOW RELATIVE PROGRESS
- TO DO THIS, A CALCULATED FIELD HAS TO BE CREATED
- IN THIS CASE, THE **PERCENTAGE DIFFERENCE** IS CALCULATED





GAUGE (PERCENTAGE)



- ONCE THE PERCENTAGE DIFFERENCE HAS BEEN CREATED, IT NEEDS TO BE MODIFIED
- PRESS THE THREE DOTS NEXT TO THE COLUMN WHICH WAS CREATED AND GO TO PROPERTIES FROM THE DROPDOWN MENU
- THE VARIABLES NEED TO BE RENAMED SO THAT THEY ARE EASIER TO UNDERSTAND AND REFER TO
- GET RID OF THE SUBTRACTION THAT OCCURS IN THE DIVISION AS THIS WILL NOW LOOK AT A DIRECT **PROFIT** AND **REVENUE** RATIO

```
1 Revenue % difference from Profit =
2 VAR __BASELINE_VALUE = SUM('Joins - 1 Key - Table 1'[Profit])
3 VAR __VALUE_TO_COMPARE = SUM('Joins - 1 Key - Table 1'[Revenue])
4 RETURN
5     IF(
6         NOT ISBLANK(__VALUE_TO_COMPARE),
7         DIVIDE(__VALUE_TO_COMPARE - __BASELINE_VALUE, __BASELINE_VALUE)
8     )
```

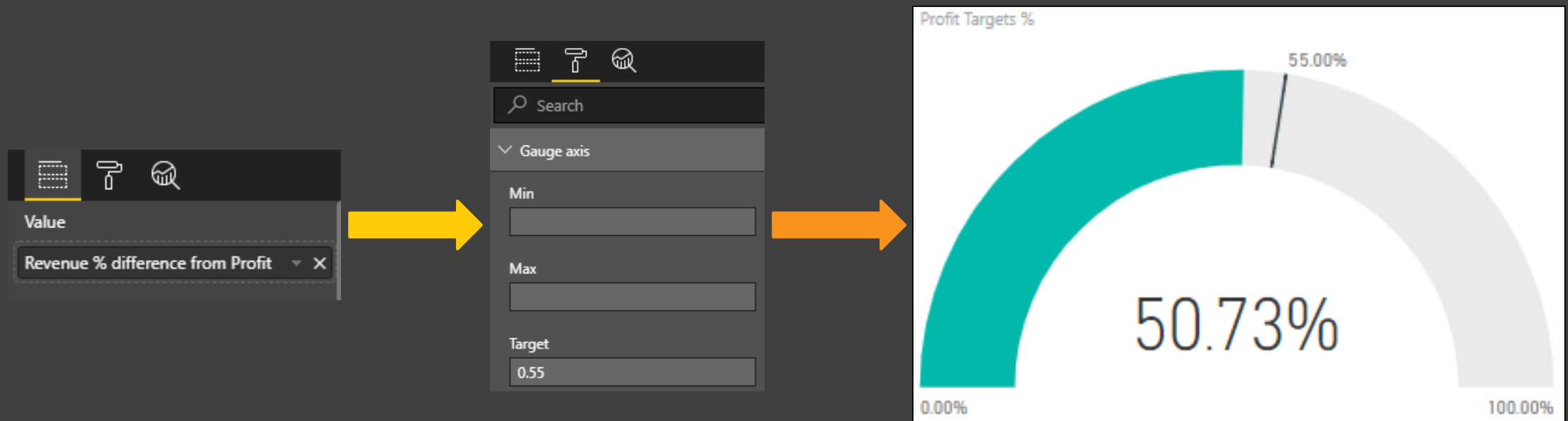


```
1 Revenue % difference from Profit =
2 VAR Profit = SUM('Joins - 1 Key - Table 1'[Profit])
3 VAR Revenue = SUM('Joins - 1 Key - Table 1'[Revenue])
4 RETURN
5     IF(
6         NOT ISBLANK(Revenue),
7         DIVIDE(Profit, Revenue)
8     )
```



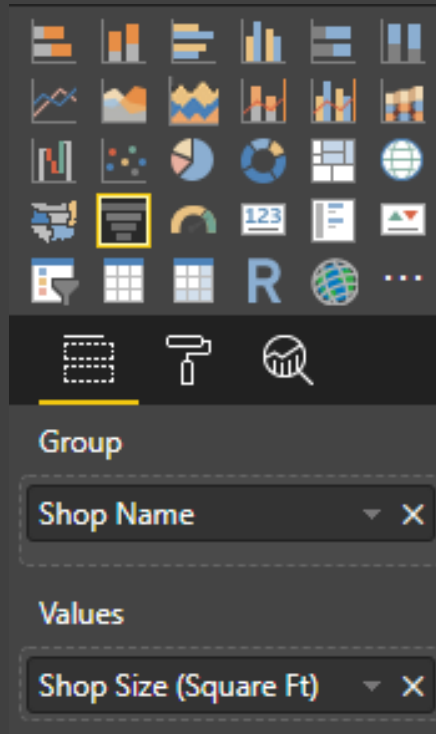
GAUGE (PERCENTAGE)

- POPULATE THE VALUE SECTION WITH THE CALCULATED PERCENTAGE COLUMN WHICH WE CREATED
- THE TARGET CAN BE DEFINED DIRECTLY AS WE ARE LOOKING AT A 55% TARGET AND WE ARE LOOKING AT THE TARGETS BASED ON PERCENTAGES

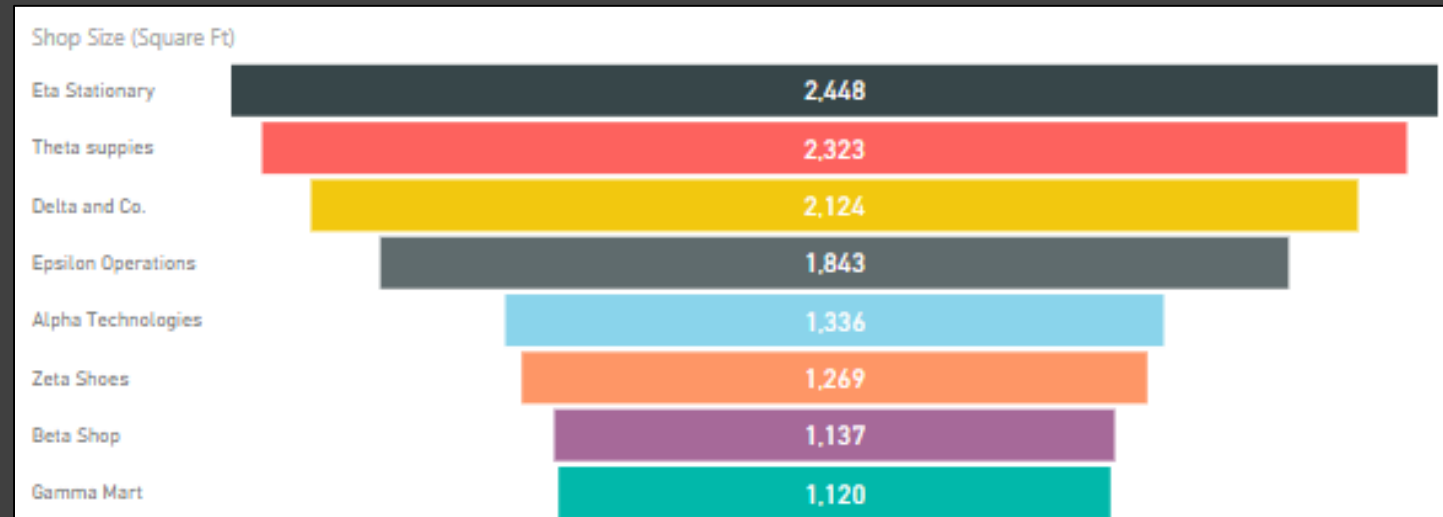




FUNNEL CHART

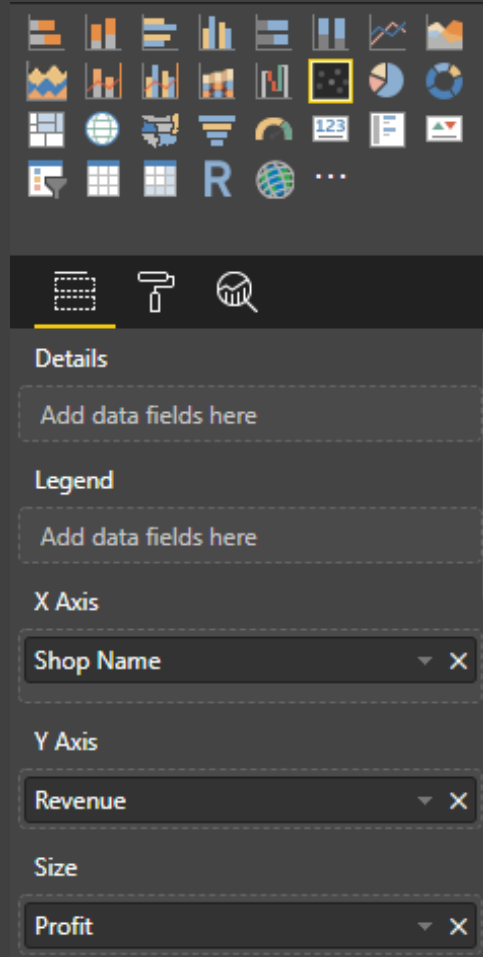


- THE FUNNEL CHART ALLOWS US TO SEE HOW EACH UNIQUE ENTRY IN THE CATEGORY FIELD IS PERFORMING AGAINST A METRIC
- THE 'GROUP' SECTION IS WHERE THE CATEGORY FIELD IS DEFINED (**SHOP NAME**)
- THE 'VALUES' SECTION IS WHERE THE METRIC OF INTEREST IS DEFINED (**SHOP SIZE**)





SCATTER CHART



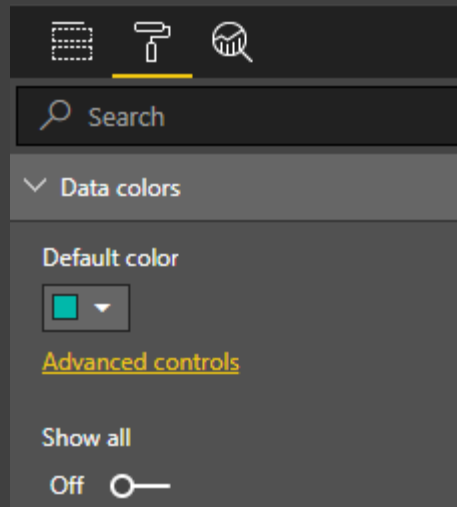
- THE SCATTER CHART ALLOWS US TO COMPARE TWO METRICS FOR THE UNIQUE ENTRIES IN A CATEGORY FIELD
- THE 'X AXIS' SECTION IS WHERE LIST THE CATEGORY FIELD IS DEFINED (**SHOP NAME**)
- THE 'Y AXIS' SECTIONS IS WHERE THE FIRST METRIC IS DEFINED (**REVENUE**)
- THE 'SIZE' IS WHERE THERE SECOND METRIC IS DEFINED (**PROFIT**). THE BIGGER THE PROFIT, THE BIGGER THE SIZE OF THE BUBBLE





SCATTER CHART – BUBBLE COLOURS

- TO CHANGE THE COLOURS OF THE BUBBLES, THE ADVANCED CONTROLS HAVE TO BE CONFIGURED IN THE 'DATA COLOURS' SECTION
- THE COLOUR SATURATION OF THE BUBBLES CAN BE CONFIGURED BASED ON THE DESIRED VALUE



Default color - *Data colors*

Format by Color scale ▼ [Learn more](#)

Based on field Sum of No of Employees ▼	Summarization Sum ▼	Default formatting ⓘ As zero ▼
Minimum Lowest value ▼ (Lowest value)	Center Middle value ▼ (Middle value)	Maximum Highest value ▼ (Highest value)

☒ Diverging

10. BEYOND THE COURSE



CONGRATULATIONS!

- **CONGRATULATIONS ON REACHING THE END OF THIS COURSE! YOU NOW HAVE THE KNOWLEDGE AND FOUNDATION TO BE AN AMAZING DATA ANALYST!**
- REMEMBER TO KEEP ON PRACTISING, AND FAMILIARISE YOURSELF WITH HOW EACH VISUALISATION CAN BE UTILISED
- IF YOU DO COME ACROSS ANY ISSUES, THEN REMEMBER TO UTILISE GOOGLE AND FORUMS TO HELP YOU WITH MAXIMISING YOUR UNDERSTANDING ON POWER BI
- YOU CAN NOW ANALYSE DATA IN SQL AND PROCESS THE RESULTS IN POWER BI TO PRODUCE DASHBOARDS! THIS WILL MAKE YOU A VERSATILE DATA ANALYST